

SeatSwap

Build specification for AI coding platforms and developers

Mobile-first PWA · Indian trains · Razorpay + PayPal

September 2026

AGENTS.md — SeatSwap build rules for AI coding agents

Read this file first, then docs/01-PRD.md ... docs/10-BUILD-PLAN.md in order. The screen images in designs/ are the visual reference for layout and wording.

Product in one line

SeatSwap is a mobile-first PWA that helps passengers on the same Indian train, same date, swap berths/seats with each other by mutual agreement.

Non-negotiable rules (never break, never "improve")

1. **Price:** ₹99 per swap = ₹49 SeatSwap fee + ₹50 thank-you credit to the acceptor. No discounts (no senior/women/first-time discounts). No coupons.
2. **Pay only after acceptance.** Sending requests is free. Requester pays ₹99 after someone accepts. No payment timer. Swap locks the moment payment succeeds; until then other matches can still accept.
3. **Acceptor never pays.** Acceptor earns ₹50 credit only when the swap is confirmed as done.
4. **Credit is never cash.** Credit only lowers future SeatSwap fees. Cannot be withdrawn or transferred (exception: can be used on a linked family trip). Expires 12 months after it is earned.
5. **No free rewards.** Nothing for signing up, adding a PNR, or being "open to swap". No "thanks for trying" credit.
6. **Swap didn't happen → ₹99 goes to the requester's credit,** never back to the bank. Only exception: a payment that failed at the bank/gateway level (money auto-returned by bank in 3-5 days).
7. **No promised dispute time.** Copy: "We'll look at both sides and reply as soon as we can. Your money is held safely meanwhile."
8. **Sign-in: Google only.** No phone OTP, SMS, WhatsApp OTP, email/password. Sign-in is asked only when the user first sends or accepts a request (PNR entry works signed out).
9. **Payments:** Razorpay for Indian payers (UPI, cards, net banking, wallets). PayPal for international travellers (label: "International traveller? Pay with PayPal"). Amount is always ₹99 INR.
10. **Banned words anywhere in UI/copy/notifications:** "TTE", "pass", "swap pass", "official", "Indian Railways", "IRCTC approved", "authorised", "legal", "grievance". Use **"Swap summary"** for the post-payment card.
11. **Positioning:** SeatSwap is a convenience platform. Every swap is the passengers' own mutual understanding. Footer line: "SeatSwap is not an official railway service." No legal/grievance screens, no lawyer talk.
12. **Navigation:** exactly 3 bottom tabs: **Home / Swaps / Profile**. Setup screens (language, privacy, alerts, Google sign-in, WhatsApp link landing) show no tab bar.
13. **Privacy:** never show another user's PNR, full name, phone, email, or ticket photo. Before payment, other users see only first name + initial, class, coach, berth type. Exact berth numbers are revealed to both sides only after payment.
14. **Scope:** Indian trains only (all trains, all India). Anyone worldwide may pay via PayPal. Buses/flights are out of scope.
15. **Low running cost:** no SMS, no masked calling, no paid OTP. Notifications = free web push + in-app Updates list.

Tech stack (fixed)

- TanStack Start v1 (React 19, Vite 7), TypeScript strict, Tailwind CSS v4, shadcn/ui.
- Backend: Supabase-compatible Postgres + Auth (Google) + RLS + Storage (Lovable Cloud).
- Server logic: createServerFn; webhooks at /api/public/* with signature verification.
- PWA: manifest + vite-plugin-pwa (generateSW, NetworkFirst for pages) for offline Trips and Swap summary.
- Roles in a separate user_roles table with has_role() security-definer function. Never check admin on the client.

Coding conventions

- Money stored as integer paise. ₹99 = 9900.
- All state transitions go through server functions that write an activity_log row.
- Every table: RLS on, explicit GRANTS.
- All copy in i18n files (/locales/{lang}.json). English + Hindi first; structure ready for all 22 scheduled languages.
- Semantic Tailwind tokens only (no raw hex in components).

01 — Product Requirements (PRD)

Problem

Indian train passengers often get berths that don't suit them: families split across coaches, elders on upper berths, groups scattered. Today they ask strangers on board, awkwardly and with no way to find who wants to swap. SeatSwap finds the right person on the same train in advance or on board.

Users

Role	Who	Goal
Requester	Passenger wanting a better berth (family, elder, medical, group)	Find a willing person fast; pay only if someone says yes
Acceptor	Passenger open to swapping	Help someone, earn ₹50 credit, stay safe
Organiser	One person managing a family/wedding/pilgrim group	Seat everyone together
Admin	SeatSwap team (1 person at launch)	See every action, fix problems, move money to credit

Core value

"Tell us what berth you want. We find people on your train who want to swap. You pay ₹99 only when someone says yes."

Scope v1

- All Indian trains, all classes (1A, 2A, 3A, 3E, SL, CC, EC, 2S).
- Add PNR (type or paste IRCTC booking SMS; parse locally). Automatic PNR lookup is a later phase.
- Multiple PNRs per user; multiple passengers per PNR; passengers in different coaches.
- Family/group trips: link PNRs; organiser pays; partial results ("3 of 4 together"). Group price ₹199 covers up to 3 swaps.
- Ranked preferences: 1st, 2nd, 3rd choice of berth type.
- Send to many matches for free; first to accept wins; requester pays to lock.
- Before-chart and on-board swaps (live coach board until your stop).
- Chat inside the app (text + quick replies; no attachments, no calls).
- Both-sides confirmation "Did you swap?".
- Credit wallet, receipts, payment history.
- Sharing: WhatsApp, Instagram, Facebook, Telegram, SMS, Copy link, QR.
- Public train pages (berth layout) for search traffic.
- Admin console: activity log, users, swaps, payments, credits, reports, overview.
- Languages: English + Hindi translated at launch; i18n ready for all 22.
- Easy mode (bigger text, one button), "Swapping for my parents".

Out of scope v1

Buses, flights, trains outside India, cash payouts, phone OTP, masked calls, SMS alerts, automatic PNR lookup, legal/grievance pages.

Success metrics (track from day 1)

PNRs added · requests sent · acceptances · paid swaps · confirmed swaps · "you're the first on this train" rate · shares per swap · repeat trips · credit redeemed.

Growth loops (must ship in v1)

1. **Never a dead end:** zero matches → "You're the first on this train" + share coach link.
2. **Accept from a link:** a shared link opens in the browser; recipient signs in with Google and accepts without installing.

3. **Coach board link** for WhatsApp groups (train + date).
4. **Chart-is-out push:** "Chart's out. 4 people in your coach want to swap."
5. **Share cards** after a swap (no private details) for status/stories.
6. **Acceptor nudge:** "You earned ₹50. Family travelling soon? Add their PNR."
7. **Welcome back** screen leading with waiting credit.
8. **Free berth check** by PNR (useful even without swapping).

02 — Data Model (Postgres)

Money = integer paise. All timestamps timestamptz. RLS enabled on every table with explicit GRANTS.

```
create type app_role as enum ('admin','support','user');
create type travel_class as enum ('1A','2A','3A','3E','SL','CC','EC','2S');
create type berth_type as enum ('LB','MB','UB','SL','SU','WINDOW','AISLE','MIDDLE_SEAT');
create type ticket_status as enum ('CNF','RAC','WL','CAN');
create type quota as enum ('GN','SS','LD','HP','TQ','PT','OTHER'); -- SS senior, LD ladies, HP disability
create type request_status as enum ('draft','searching','accepted_awaiting_payment','locked','confirmed','voided','disputed','expired','withdrawn');
create type offer_status as enum ('sent','accepted','declined','superseded','expired');
create type outcome as enum ('swapped','no_show','not_possible','changed_mind');
create type pay_provider as enum ('razorpay','paypal','credit');
create type pay_status as enum ('created','pending','paid','failed');

profiles(id uuid pk -> auth.users, first_name, last_initial, gender_optional, language, easy_mode bool,
rating numeric, created_at)
user_roles(id, user_id -> auth.users, role app_role, unique(user_id, role))
settings(user_id pk, women_only bool, families_only bool, same_coach_only bool, paused bool,
max_requests_per_day int default 3, notify_push bool)

bookings(id, user_id, pnr_hash text, pnr_last4 text, train_no, train_name, journey_date date,
from_code, to_code, class travel_class, is_chair_car bool, source ('typed'|'sms_paste'),
chart_prepared bool, created_at) -- never store full PNR in plain text
passengers(id, booking_id, label ('Passenger 1'), coach, berth_no, berth_type, status ticket_status,
quota quota, is_child_no_berth bool, board_code, drop_code)

group_trips(id, organiser_id, name, created_at)
group_members(group_id, booking_id)

swap_requests(id, requester_id, booking_id, passenger_ids uuid[], group_id null,
choices berth_type[3], -- ranked 1st/2nd/3rd
same_coach bool, keep_together bool, reason text, status request_status,
locked_offer_id null, created_at, updated_at)
swap_offers(id, request_id, acceptor_id, acceptor_booking_id, acceptor_passenger_id,
matched_choice_rank int, status offer_status, created_at, responded_at)

payments(id, request_id, payer_id, provider pay_provider, provider_ref, amount_paise int,
credit_used_paise int, currency 'INR', status pay_status, created_at)
receipts(id, payment_id, number 'SS-10482', pdf_path)

wallet_tx(id, user_id, amount_paise int, kind ('acceptor_credit'|'swap_to_credit'|'used'|'expired'|'admin_adjust'),
ref_request_id, expires_at, created_at) -- balance = sum(amount) where not expired

confirmations(request_id, user_id, outcome outcome, created_at, primary key(request_id,user_id))
disputes(id, request_id, status ('open'|'resolved'), resolution, admin_id, created_at)

chats(id, request_id) ; messages(id, chat_id, sender_id, text, flagged_risky bool, created_at)
reports(id, reporter_id, reported_id, request_id, reason, status ('open'|'closed'), created_at)
blocks(blocker_id, blocked_id)

notifications(id, user_id, kind, title, body, link, read_at, created_at) -- in-app Updates list
push_subscriptions(user_id, endpoint, keys jsonb)

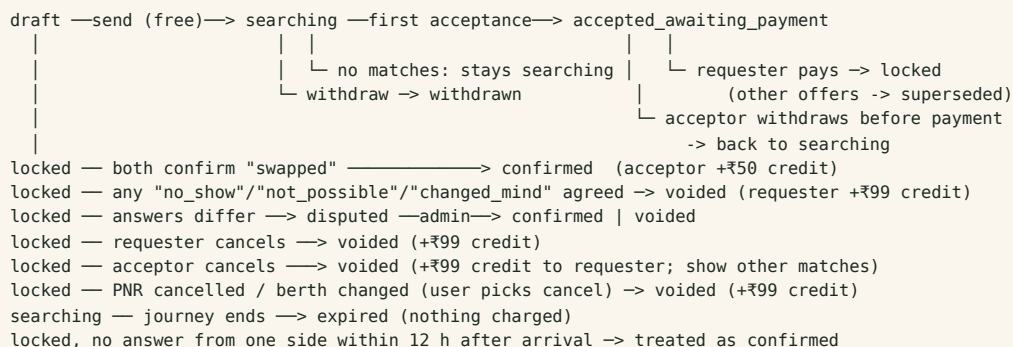
activity_log(id, actor_id, actor_role, action text, entity text, entity_id uuid,
meta jsonb, created_at) -- EVERY state change writes here
```

Visibility rules (enforce in RLS + server functions)

- Other users' PNR, names beyond first name + initial, phone, email: never readable.
- passengers.berth_no of the other party readable only when the request is locked or later.
- activity_log, reports, disputes: admin/support only (via has_role).
- Users read only their own wallet_tx, payments, receipts, notifications.

03 — State Machines

Swap request



Offer

sent → accepted | declined | expired; when another offer on the same request gets paid, all open/accepted offers → superseded and those acceptors see "Someone else was faster".

Payment

created → pending → paid | failed. failed = nothing charged (bank returns any debit in 3-5 days). Only paid locks the swap. Webhook is the source of truth; client polling only updates UI.

Money outcomes table

Situation	Requester	Acceptor	SeatSwap
Swap confirmed	paid ₹99	+₹50 credit	keeps ₹49
Not possible / no-show / changed mind	+₹99 credit	nothing	nothing
Requester cancels after paying	+₹99 credit	nothing	nothing
Acceptor cancels after payment	+₹99 credit	nothing (counts toward backing-out limit)	nothing
Nobody accepts	₹0 charged	–	–
Payment failed at bank	₹0 (bank auto-return)	–	–
Dispute	held until admin decides	held	–

Credit used on a payment: up to the full ₹99 may be covered by credit; provider = 'credit' when fully covered.

Abuse limits

- Backed out 3 times in 30 days → hidden from matches for 30 days.
- Max outgoing requests per user per day: 10. Acceptor setting `max_requests_per_day` (default 3).
- One berth can be in only one locked swap at a time.

04 — User Flows with If/Else Rules

A. Requester

1. **First open** → "Where's your train taking you?" PNR box, "Paste from IRCTC SMS", line "Earn ₹50 credit every time you help someone swap". No sign-in.
2. **Language** (first open only) → English / हिन्दी / more.
3. **Before you start** (once): convenience app · swap only if both agree · keep your original ticket and ID.
4. **Add PNR** → parse → **Your berth** card (coach, berth type, "3 people on this train want to swap").
 - PNR invalid → "Check the 10 digits" + retry.
 - WL/RAC → show ticket, "Swaps open once your berth is confirmed", notify on confirm.
 - CAN → "This ticket is cancelled", cannot swap.
 - Quota berth (SS/LD/HP) → soft note: "This berth was given for a reason; it's only offered to people who qualify."
 - Child without berth → counted in group, never offered.
 - Chair car (CC/EC/2S) → use seat words: Window / Aisle / Middle, facing direction.
5. **What would you like?** Rank 1st/2nd/3rd choice, same coach toggle, keep group together, reason chips.
6. **Matches** → same train, same date, same class, overlapping journey, offers a berth in the choices. Berth shown as "Berth ••". Tick many → "Send to N · free".
 - 0 matches → **You're the first on this train** → share coach link (all platforms).
7. **Google sign-in** (first send only) → **Privacy** consent (alerts unticked) → **Turn on alerts** (Not now allowed).
8. **Your request** (manage): edit, pause, withdraw. No reply → "Share" / "Try another berth type".
9. **Someone says yes** (push + Updates) → **Pay ₹99** screen: breakdown ₹49 + ₹50, credit line only if balance > 0, "No swap? ₹99 goes to your credit".
 - 2nd-choice acceptance → "You got your 2nd choice" · Accept this / Keep waiting (1st choice stays open until paid).
 - Two accept at once → first to be paid for wins; others see "Someone else was faster".
10. **Choose how to pay:** Razorpay (GPay, PhonePe, Paytm, UPI ID, card, net banking) or "International traveller? Pay with PayPal".
 - UPI waiting → approve in app. Pending → "Please don't pay again", Check status. Failed → "No money was taken", Try again / Pay another way.
11. **Payment successful** → receipt → berth numbers revealed → chat opens.
12. **On board:** chat (quick replies "I'm at my berth now", "Meet me near the coach door"), cash-word warning + Report & block, **Found each other?** ("I've met Arjun" / "Can't find them"), **Swap summary** (works offline).
13. **Did you swap?** Yes / Other person didn't come / Swap wasn't possible / We changed our mind.
14. **After:** "You swapped" → share card → rating → add another trip.
 - Went wrong → ₹99 to credit (valid 12 months). Answers differ → held, reply as soon as we can.
15. **Changes:** PNR cancelled → ₹99 to credit. Berth changed after chart → Keep swap / Cancel · move to credit.

B. Acceptor

1. Adds PNR → "I'm open to swap" (no reward for this).
2. Filters (Settings): women only, families only, same coach, pause, max per day.
3. Request arrives (push / Updates / WhatsApp link / on-board board): shows "You give Lower · you get Upper", first name, reason, "Earn ₹50 credit · you pay nothing". Accept / Decline.
 - From link without app → browser page, Google sign-in, accept.
4. **Waiting for payment:** "Priya pays ₹99. You pay nothing." Can back out until paid.
 - Someone else was faster → friendly message, stay open.
5. After payment → berth revealed, chat, Found each other, Swap summary, Did you swap?
6. Confirmed → **You earned ₹50** → share good deed → rating → "Travelling again? Add PNR".
7. Can back out any time before physically moving; counts toward the limit after payment.

C. Organiser / family

Link PNRs → see everyone on one coach map → "Seat everyone together" → plan finds swaps → partial result "3 of 4 together" → pay ₹199 once (up to 3 swaps). "Swapping for my parents": organiser handles chat; parents get the Swap summary link.

D. Admin

Overview (today's PNRs, requests, paid, confirmed, credit issued, busiest trains) → Activity log (every action by every user; filter by action, search by user/PNR last4/train, export CSV) → Users (search, reported/blocked, history, block) → Swaps (by status, timeline, Move to credit, Mark done, resolve dispute) → Payments (Razorpay/PayPal, paid/failed, moved to credit) → Credits (issued, used, expiring, manual adjust with reason) → Reports (review, close, block). Every admin action is itself logged.

Notifications (web push + in-app Updates)

someone accepted · payment done · chart is out · new request (acceptor) · partner arrived message · did you swap? (after arrival) · credit added · credit expiring in 30 days · request expired.

05 — Screen List & Routes

Tab = which bottom tab is highlighted. "—" = no tab bar. Design page refers to designs/NN *.jpg (3 screens per page, left→right = a/b/c).

#	Screen	Route	Tab	Design
1	First open (empty home)	/	Home	25a
2	Choose language	/welcome/language	—	10a
3	Before you start	/welcome/note	—	12a
4	Home · my trips	/	Home	1a
5	Add PNR	/trips/add	Home	1b
6	Your berth	/trips/\$id	Home	1c
7	Waitlisted / RAC	/trips/\$id	Home	8c
8	Quota berth note	/trips/\$id	Home	9a
9	Chair car seat picker	/request/new	Swaps	8b
10	Rank your choices	/request/new	Swaps	14a
11	Matches · send free	/request/\$id/matches	Swaps	2a
12	Google sign-in	/signin	—	3b
13	Privacy	/welcome/privacy	—	11b
14	Turn on alerts	/welcome/alerts	—	10c
15	Your request (manage)	/request/\$id	Swaps	12b
16	No reply yet	/request/\$id	Swaps	12c
17	You're the first	/request/\$id	Swaps	7b
18	Invite / share anywhere	/share/\$trainDate	Swaps	14b
19	Chart is out	push → /trips/\$id	Home	7c
20	Updates	/updates	Swaps	13a
21	Arjun said yes · pay ₹99	/pay/\$requestId	Swaps	3c
22	2nd choice match	/request/\$id	Swaps	14c
23	Choose how to pay	/pay/\$requestId/method	Swaps	27a
24	Waiting for UPI	/pay/\$requestId/upi	Swaps	27b
25	Pay with PayPal	/pay/\$requestId/paypal	Swaps	28a
26	Payment pending	/pay/\$requestId/status	Swaps	28b
27	Payment failed	/pay/\$requestId/status	Swaps	28c
28	Payment successful	/pay/\$requestId/done	Swaps	27c
29	Someone else was faster	/swaps/\$id	Swaps	20a
30	Partner cancelled	/swaps/\$id	Swaps	25b
31	Cancel this swap?	/swaps/\$id/cancel	Swaps	25c
32	Swap request (acceptor)	/incoming/\$id	Swaps	4a

#	Screen	Route	Tab	Design
33	Accept from link (browser)	/s/\$code	—	8a
34	On the train now board	/onboard/\$tripId	Home	7a
35	Waiting for payment (acceptor)	/incoming/\$id	Swaps	19c
36	Chat	/chat/\$id	Swaps	4b
37	Cash warning	/chat/\$id	Swaps	11c
38	Found each other?	/swaps/\$id/meet	Swaps	13b
39	Swap summary (offline)	/swaps/\$id/summary	Swaps	9c
40	Did you swap?	/swaps/\$id/confirm	Swaps	4c
41	You swapped	/swaps/\$id/done	Swaps	2c
42	Share your trip card	/swaps/\$id/share	Swaps	6c
43	You earned ₹50	/swaps/\$id/done	Swaps	26a
44	Share good deed	/swaps/\$id/share	Swaps	26b
45	Travelling again?	/swaps/\$id/done	Swaps	26c
46	Rating	/swaps/\$id/rate	Swaps	13c
47	Something went wrong	/swaps/\$id	Swaps	6b
48	Added to your credit	/swaps/\$id	Swaps	20b
49	Answers don't match	/swaps/\$id	Swaps	20c
50	PNR cancelled	/trips/\$id	Home	21a
51	Berth changed	/trips/\$id	Home	21b
52	Family trip	/groups/\$id	Home	19a
53	Seat everyone together	/groups/\$id/plan	Home	19b
54	Swapping for my parents / easy mode	/profile/easy	Profile	10b
55	Welcome back	/	Home	11a
56	Swaps list	/swaps	Swaps	5b
57	Profile & credit	/profile	Profile	5c
58	Payments & receipts	/profile/payments	Profile	29a
59	Receipt	/profile/payments/\$id	Profile	29b
60	Settings	/profile/settings	Profile	21c
61	Help	/profile/help	Profile	22a
62	Delete account	/profile/delete	Profile	22b
63	Account deleted	/goodbye	—	22c
64	No internet	any	current	29c
65	Public train page	/train/\$number	— (web)	-
A1-A6	Admin: Overview, Activity log, Users,	/admin/*	desktop sidebar	23, 15, 16, 17, 18, 24

#	Screen	Route	Tab	Design
	Swaps, Payments & reports, Credits			

The phone screens are mobile-first (360–430 px). The admin screens are responsive desktop pages.

06 — Payments (Razorpay + PayPal)

Choosing the provider

- Default: **Razorpay** (Indian cards, UPI, net banking, wallets). Show UPI apps first.
- "International traveller?" → **PayPal**. Also auto-suggest PayPal when the browser locale/currency is non-INR, but let the user switch.
- Amount always ₹99 INR (₹199 for a group trip). PayPal converts; show "about US\$X" as an estimate only.

Razorpay flow

1. Server fn `createRazorpayOrder(requestId)` → validates request is `accepted_awaiting_payment` and caller is the requester; computes `amount = 9900 - creditUsed`; creates order (`receipt = request id`); inserts `payments(status='created')`.
2. Client opens Razorpay Checkout with `order_id`.
3. On handler success → server fn `verifyRazorpaySignature` (HMAC SHA256 of `order_id|payment_id` with key secret) → mark pending until webhook.
4. Webhook `/api/public/webhooks/razorpay` (verify X-Razorpay-Signature): `payment.captured` → paid → lock request, supersede other offers, reveal berths, notify both, write `activity_log`. `payment.failed` → failed.
5. Idempotent on `provider_ref`.

PayPal flow

1. Server fn `createPaypalOrder(requestId)` (Orders v2, currency INR, intent CAPTURE).
2. Client: PayPal JS buttons → approve → server fn `capturePaypalOrder`.
3. Webhook `/api/public/webhooks/paypal` (verify via PayPal `verify-webhook-signature` API): `PAYMENT.CAPTURE.COMPLETED` → paid; `DENIED` → failed.

Credit

- If credit balance $\geq$ amount → no provider; `provider='credit'`, instant lock.
- Credit is consumed oldest-first (earliest expiry first).
- "Swap didn't happen" never triggers a provider refund; it inserts `wallet_tx(kind='swap_to_credit', +9900, expires_at = now()+12 months)`.

Receipts

Number format SS-####. Lines: SeatSwap fee ₹49 · Thank-you credit ₹50 · credit used (if any) · Total. Downloadable PDF + share.

Secrets (server only)

RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET, RAZORPAY_WEBHOOK_SECRET, PAYPAL_CLIENT_ID, PAYPAL_CLIENT_SECRET, PAYPAL_WEBHOOK_ID. Public: Razorpay key id and PayPal client id may be exposed to the browser.

Tax note for the owner

GST on the ₹49 fee should be confirmed with an accountant before live payments.

07 — Design System ("calm, like WhatsApp")

Principles

- One clear primary button per screen. Lots of white space. Plain lists. Big tap targets (min 48 px).
- Short sentences, everyday words. Numbers large (₹99, ₹50).
- Works on cheap Android phones: fast first load (< 200 KB JS for first screen goal), no heavy animation.

Tokens

Token	Value	Use
--primary	#1F6B45 (deep green)	buttons, active tab, ticks
--accent	#F2A33A (saffron)	credit, highlights, secondary outline buttons
--background	#FAF6EE (cream)	app background
--card	#FFFFFF	cards
--ink	#1D2A22	text
--muted	ink at 55%	secondary text
--danger	#C2410C	warnings (sparingly)
radius	16 px cards, 14 px buttons, full for chips	
shadow	very soft (0 1px 2px / 6%)	

Type

Headings: Plus Jakarta Sans 700. Body: Inter 400/500 (Devanagari fallback: Noto Sans Devanagari). Sizes: title 26, section 17, body 15, caption 12. Easy mode ×1.15.

Components

Top bar (wordmark "SeatSwap" left, settings icon right) · bottom tab bar (Home, Swaps, Profile) · trip card · berth chip ("Berth ••" before payment) · match row (initials avatar, "✓ Google verified", berth type, coach, choice rank) · primary button · outline button (saffron) · check row · price breakdown card · status timeline · chat bubbles + quick-reply chips · empty state with single action · offline banner.

Wording rules

- Use: "Swap summary", "thank-you credit", "No swap? ₹99 goes to your credit", "Pay only inside SeatSwap", "SeatSwap is not an official railway service."
- Never: TTE, pass, official, Indian Railways, authorised, legal, grievance, refund-to-bank (except bank failure).
- Before payment always "Berth ••".

Logo

Use one wordmark everywhere: "SeatSwap" in Plus Jakarta Sans 700, primary green. The design images vary slightly; follow this rule, not the images.

08 — PWA, Tech & Operations

PWA

- `manifest.webmanifest`: name "SeatSwap", short_name "SeatSwap", display standalone, theme #1F6B45, background #FAF6EE, icons 192/512 + maskable.
- Offline: vite-plugin-pwa generateSW; NetworkFirst for pages; CacheFirst for hashed assets. Cache "My trips" and every "Swap summary" for offline viewing. Queue chat messages while offline and send when back online.
- Never register the service worker in dev/preview/iframe; support `?sw=off` kill switch.
- Install prompt: gentle card after first successful PNR add ("Add SeatSwap to your home screen").

Push

Web Push (VAPID) — free. Fallback: in-app Updates list + badge on Swaps tab. Ask permission only after the user sends or accepts a first request.

Auth

Google OAuth only. On first sign-in create profiles row with first name + last initial. Anonymous (pre-sign-in) PNRs stored locally and attached to the account after sign-in.

PNR input

Type 10 digits or paste the IRCTC booking SMS; parse train no, date, class, coach, berth, status on the client with regex. Store only pnr_hash (SHA-256 + salt) and last 4 digits. Automatic PNR lookup via a licensed data provider is a later phase.

Matching (server)

Candidates = same train_no + journey_date + class, overlapping from/to segments, status CNF, not blocked, acceptor not paused and within their daily limit, filters (women only / families only / same coach) respected, berth type in requester's choices. Score: choice rank (1st 50, 2nd 35, 3rd 20) + same coach 10 + keep-together fit 10 + acceptor rating 0-10. Quota berths shown only to people who qualify.

Safety

Risky-message detector (cash, UPI IDs, phone numbers, "pay me", "sell") → warn both sides, hide the number, offer Report & block. Rate limits on requests and messages.

Admin & logs

`/admin` guarded by `has_role(auth.uid(), 'admin')` on the server. Activity log captures: sign_in, pnr_added, request_sent, offer_accepted, offer_declined, payment_created/paid/failed, swap_locked, message_flagged, confirmation, dispute_opened/resolved, credit_added/used/expired, report_created, user_blocked, admin_action. CSV export.

Scheduled jobs

- Expire requests after journey end.
- Auto-confirm locked swaps 12 h after arrival if one side didn't answer.
- Credit expiry (daily) + reminder 30 days before.
- Chart-time notifications (when chart status flips).

Analytics events

Same names as activity_log actions + share_clicked(platform), install_prompt_accepted, first_screen_viewed.

09 — Key Copy (English)

Place	Text
First open title	Where's your train taking you?
First open hint	Earn ₹50 credit every time you help someone swap
PNR paste	Paste from IRCTC SMS
Before you start	SeatSwap is a convenience app. A swap happens only if both of you agree. Keep your original ticket and ID with you.
Send button	Send to {n} · free
Pay screen	{name} said yes! · Pay ₹99
Breakdown	SeatSwap fee ₹49 · Thank-you credit for {name} ₹50
Under pay	No swap? ₹99 goes to your credit
Lock note	The swap locks once you pay. Until then, your other matches can still accept.
Accept card	You give {yours} · you get {theirs}. Earn ₹50 credit · you pay nothing
Waiting for payment	{name} pays ₹99. You pay nothing.
Faster	Someone else was faster. You're still open to swap.
You're the first	You're the first on {train}. Share this link so others on your train can join.
Swap summary footer	Keep your original ticket and ID with you. SeatSwap is not an official railway service.
Did you swap?	Yes, swapped · Other person didn't come · Swap wasn't possible · We changed our mind
To credit	₹99 added to your credit · valid for 12 months · never cash
Dispute	We'll look at both sides and reply as soon as we can. Your money is held safely meanwhile.
Cash warning	Pay only inside SeatSwap. Never send money to anyone directly.
Payment failed	No money was taken. If money left your account, your bank returns it in 3–5 days.
Pending	Your bank is still confirming ₹99. Please don't pay again.
Acceptor earned	You earned ₹50 credit. Thanks for helping {name}'s family sit together.
Help: allowed?	SeatSwap only helps passengers find each other. Any swap is your own agreement. Keep your original ticket and ID, and follow instructions from railway staff on board.

Hindi translations for all strings must be written/reviewed by a native speaker before launch.

10 — Build Plan (in order)

1. **Foundation:** TanStack Start app, design tokens, 3-tab shell, i18n (en, hi), PWA manifest.
2. **Trips:** add PNR (type/paste SMS parse), multiple PNRs, passengers, statuses (WL/RAC/CAN), quota, chair car, child-no-berth. Works signed out (local), synced after sign-in.
3. **Auth:** Google sign-in, profiles, privacy consent, user_roles + has_role.
4. **Requests & matching:** ranked choices, matches list with hidden berths, send to many free, offers, Updates list, "You're the first", share sheet (WhatsApp, Instagram, Facebook, Telegram, SMS, Copy, QR).
5. **Accepter:** open to swap, filters, incoming request, accept/decline, waiting for payment, "someone was faster".
6. **Payments:** Razorpay + PayPal + credit, webhooks, receipts, pending/failed states.
7. **Locked swap:** berth reveal, chat with risky-message guard, report/block, Found each other, Swap summary offline.
8. **Outcomes:** Did you swap?, credit rules, disputes, auto-confirm job, rating, share cards, acceptor nudge.
9. **Changes:** PNR cancelled, berth changed, partner cancelled, cancel swap.
10. **Groups:** link PNRs, family plan, ₹199 group payment, swapping for parents, easy mode.
11. **On board:** live coach board until your stop.
12. **Admin:** overview, activity log, users, swaps, payments, credits, reports, CSV export.
13. **Growth extras:** public train pages (SEO), free berth check, chart-time push, welcome back, credit reminders, install prompt.
14. **QA:** every flow in 04-FLOWS.md end to end on a 360 px Android device, offline tests, webhook replay tests.

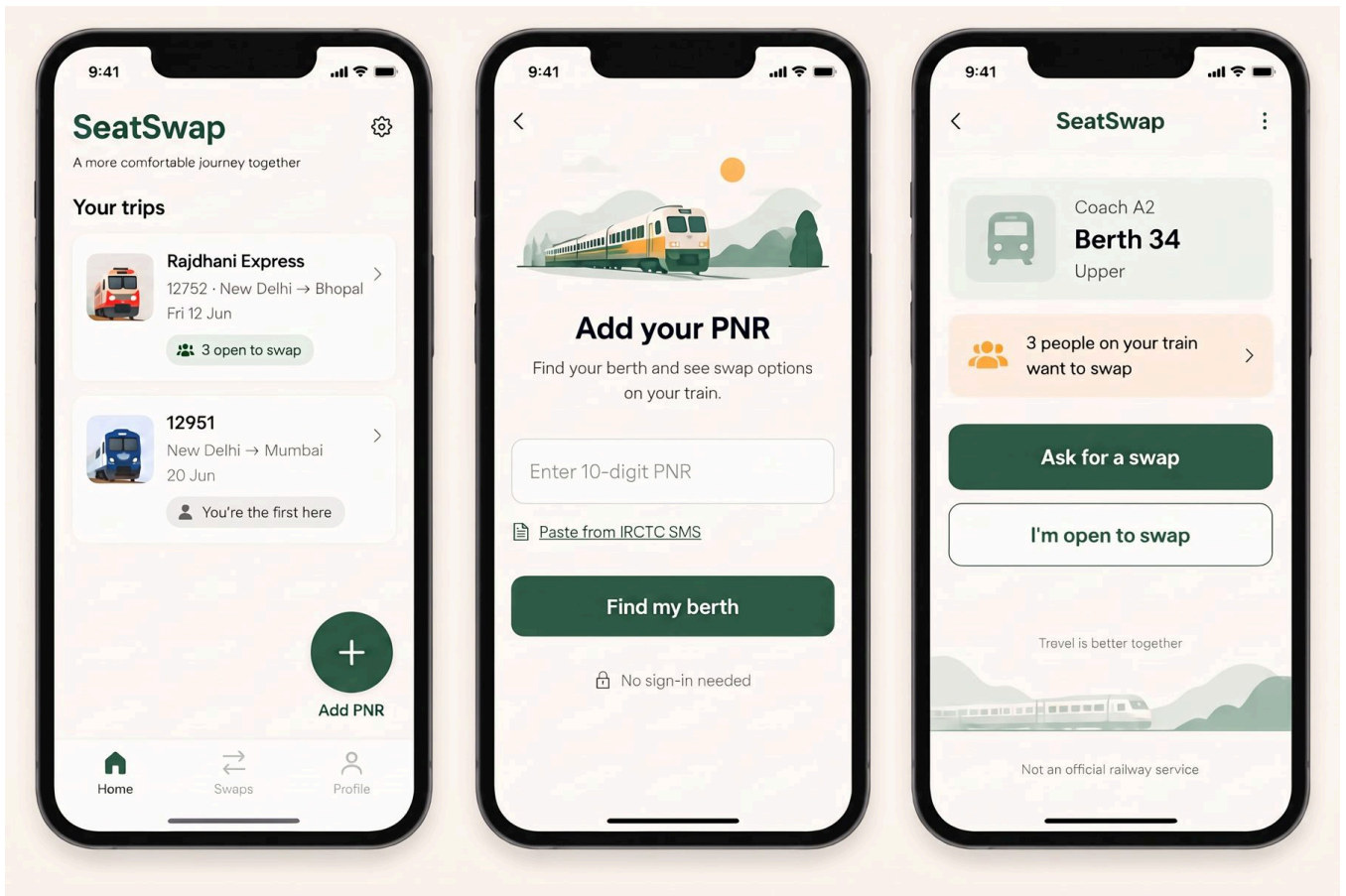
Definition of done

- All non-negotiable rules in AGENTS.md pass a manual checklist.
- No banned words anywhere (grep the locales).
- Every state change writes activity_log.
- Lighthouse PWA installable; first screen usable on slow 3G.

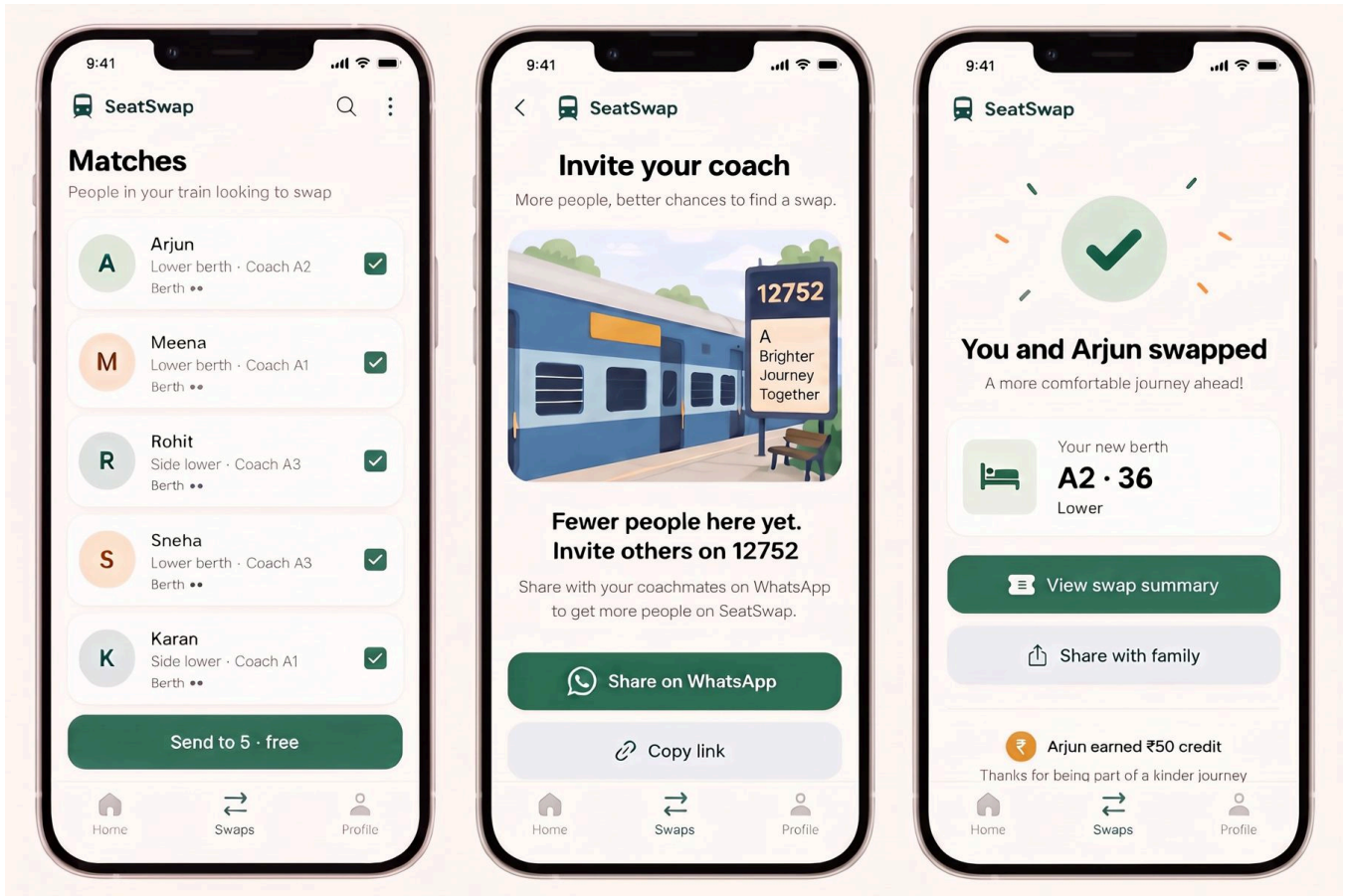
Placeholders to replace

Domain (currently planned: toyoufromme.website), example names (Arjun, Priya, Riya), admin figures, QR pattern, US\$ estimate.

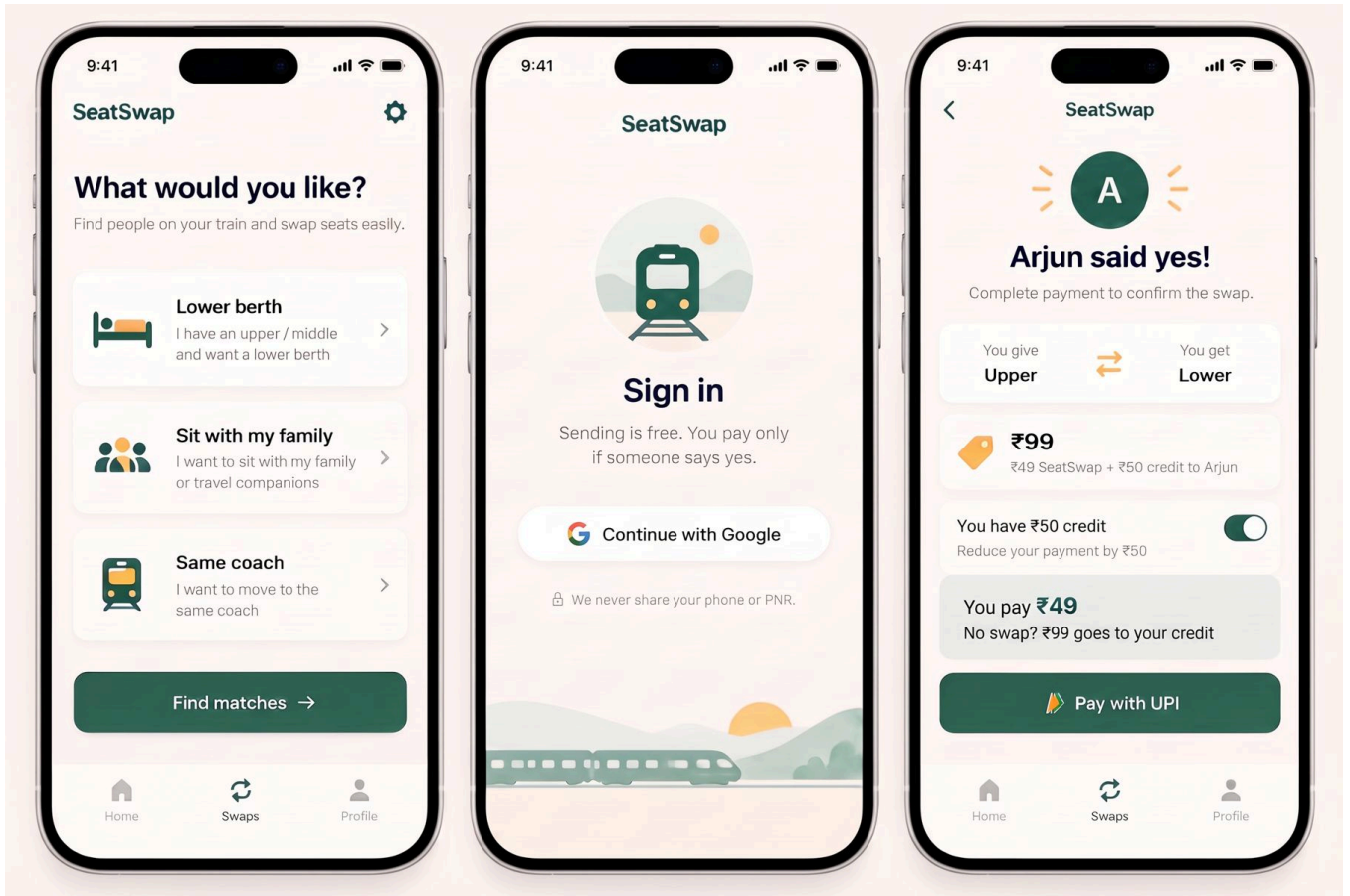
Design 01 Start



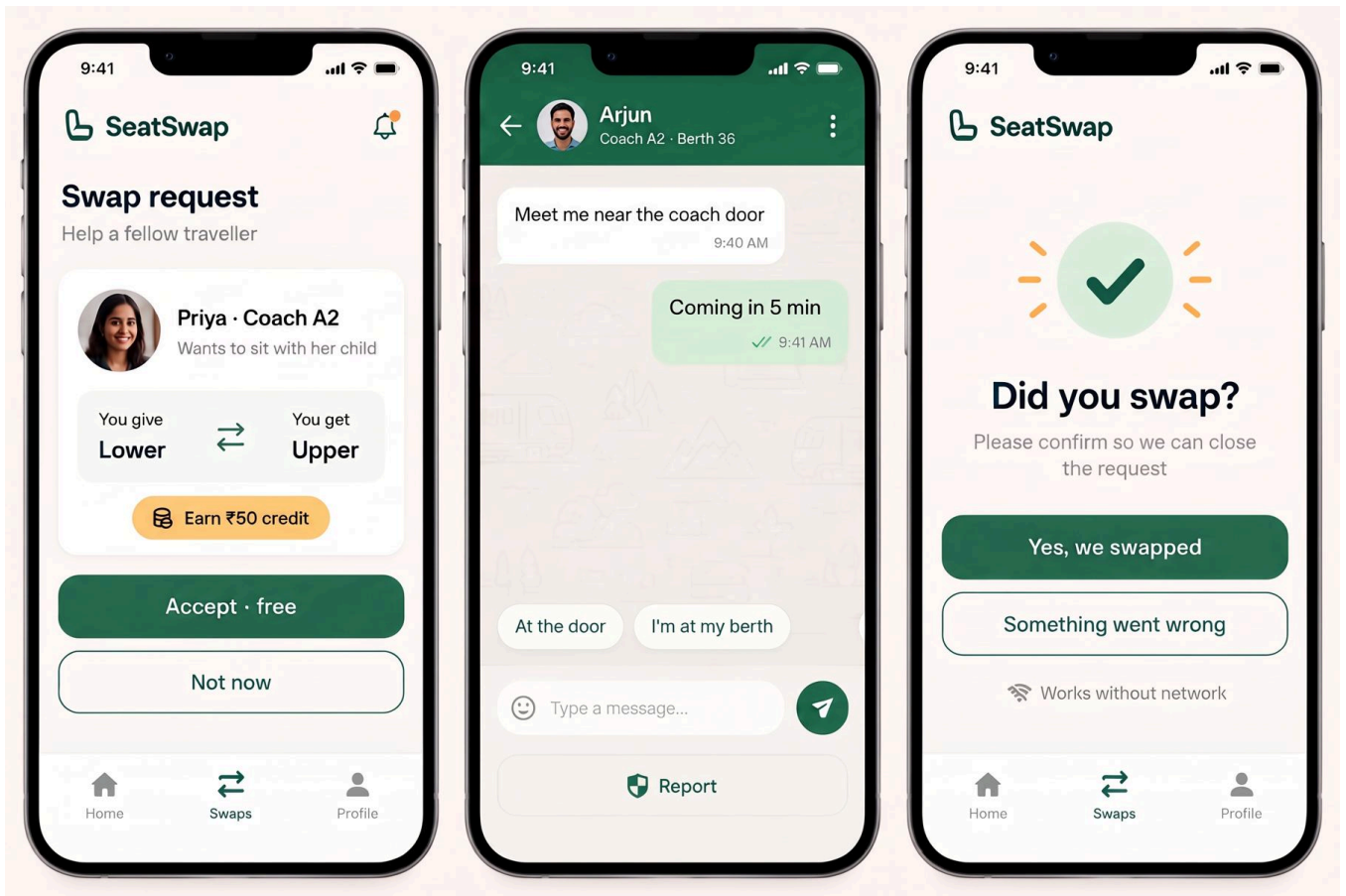
Design 02 Match and share



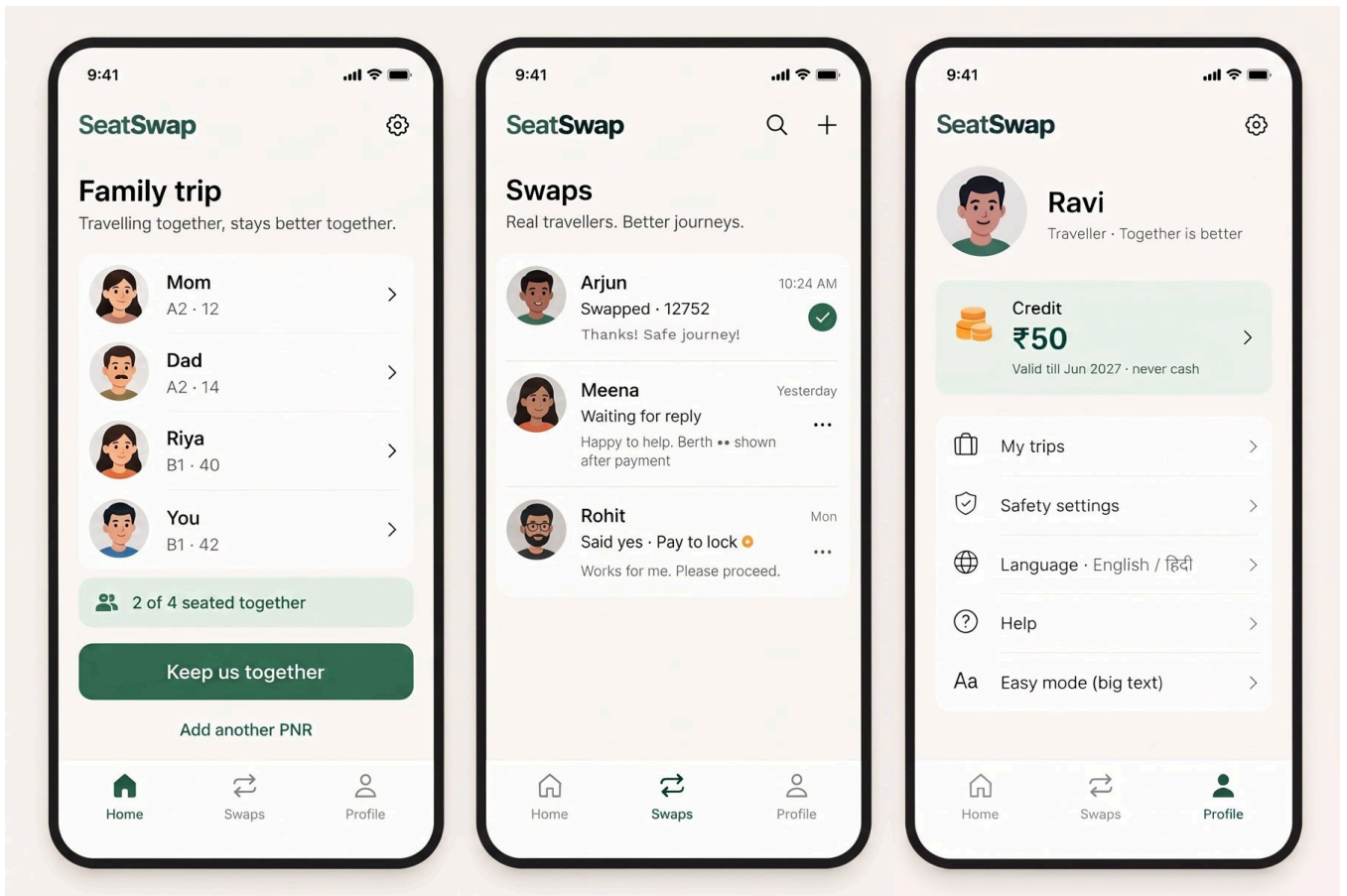
Design 03 Ask and pay



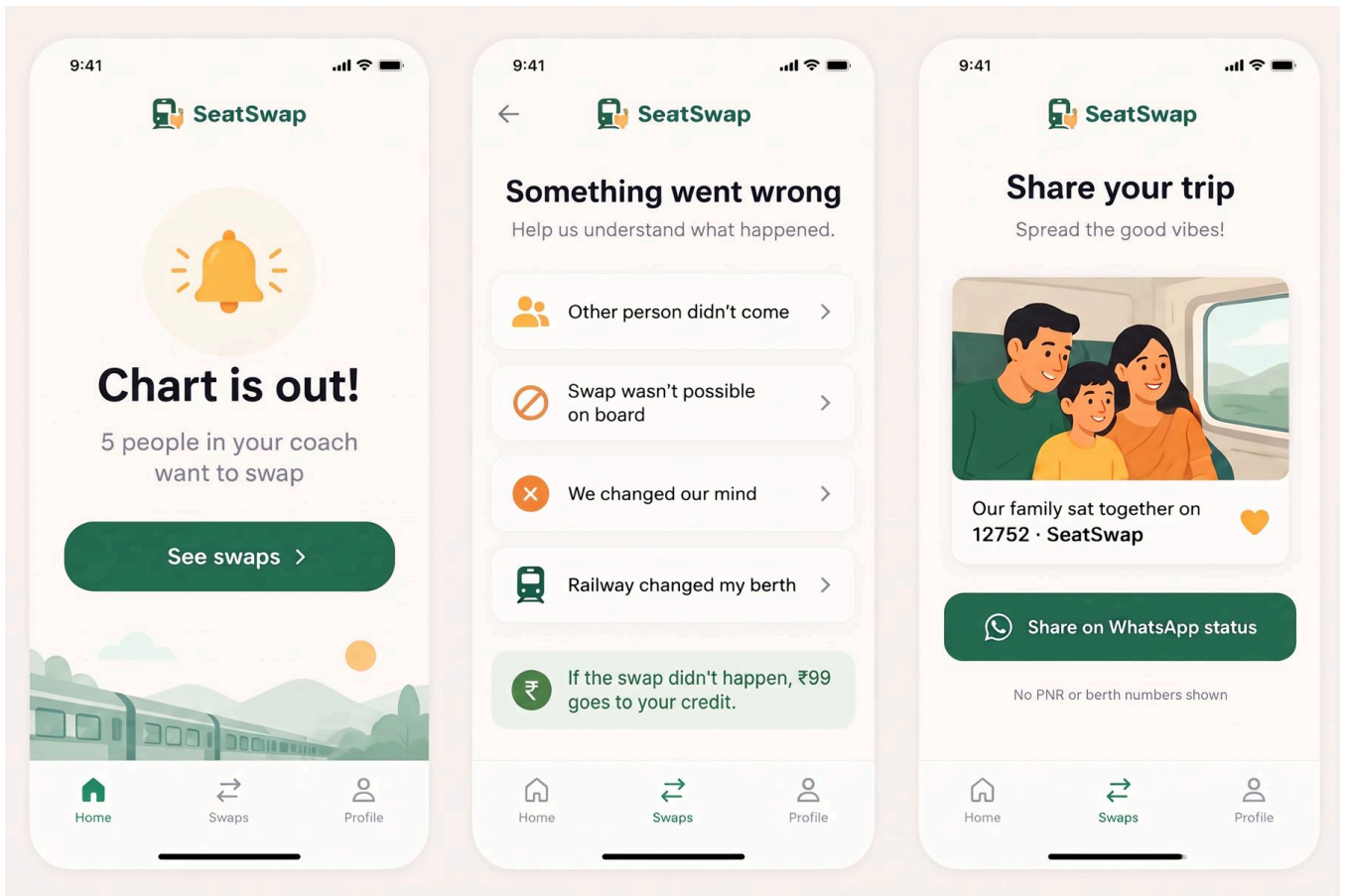
Design 04 Accept and travel



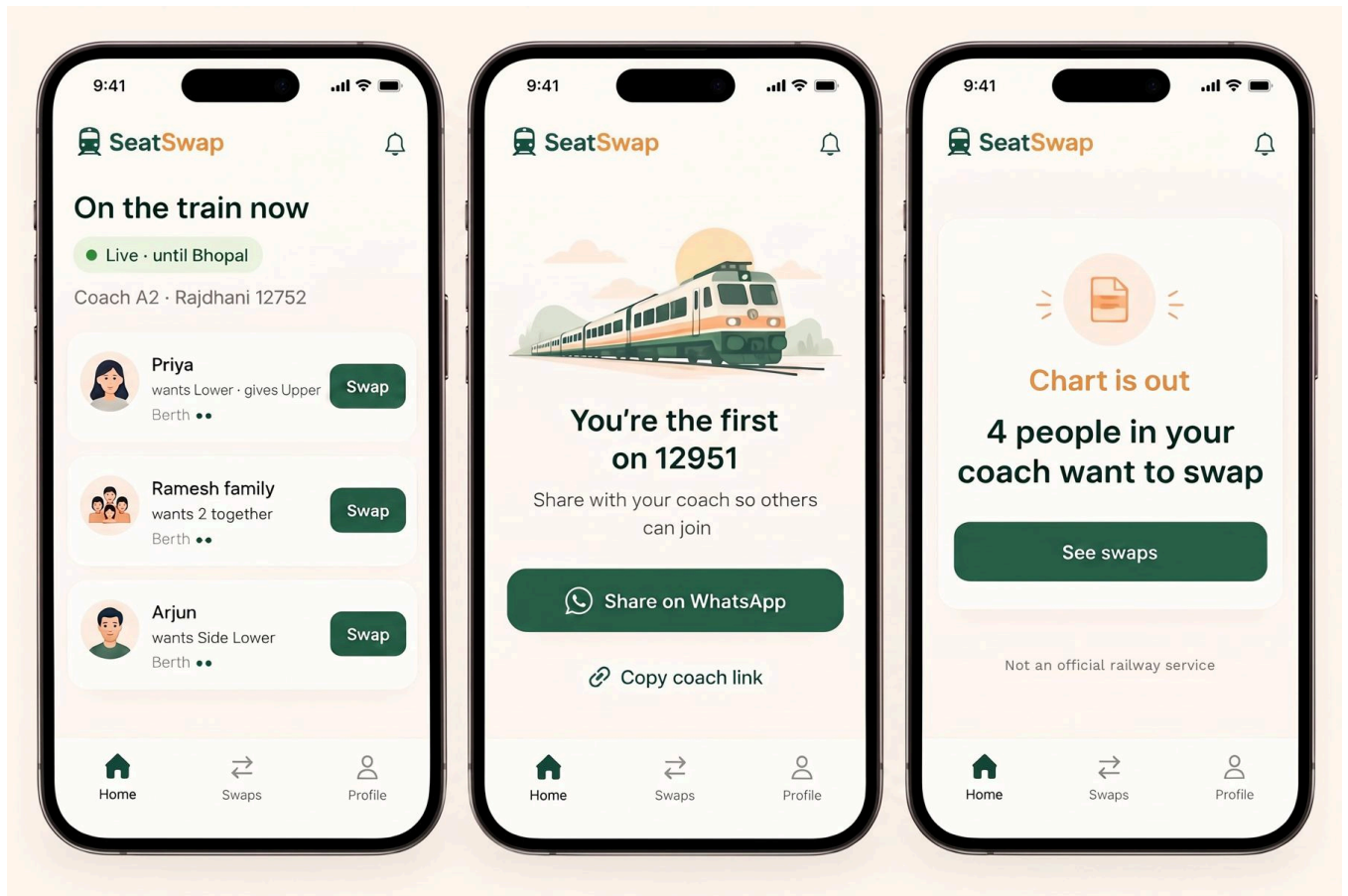
Design 05 Family, swaps, profile



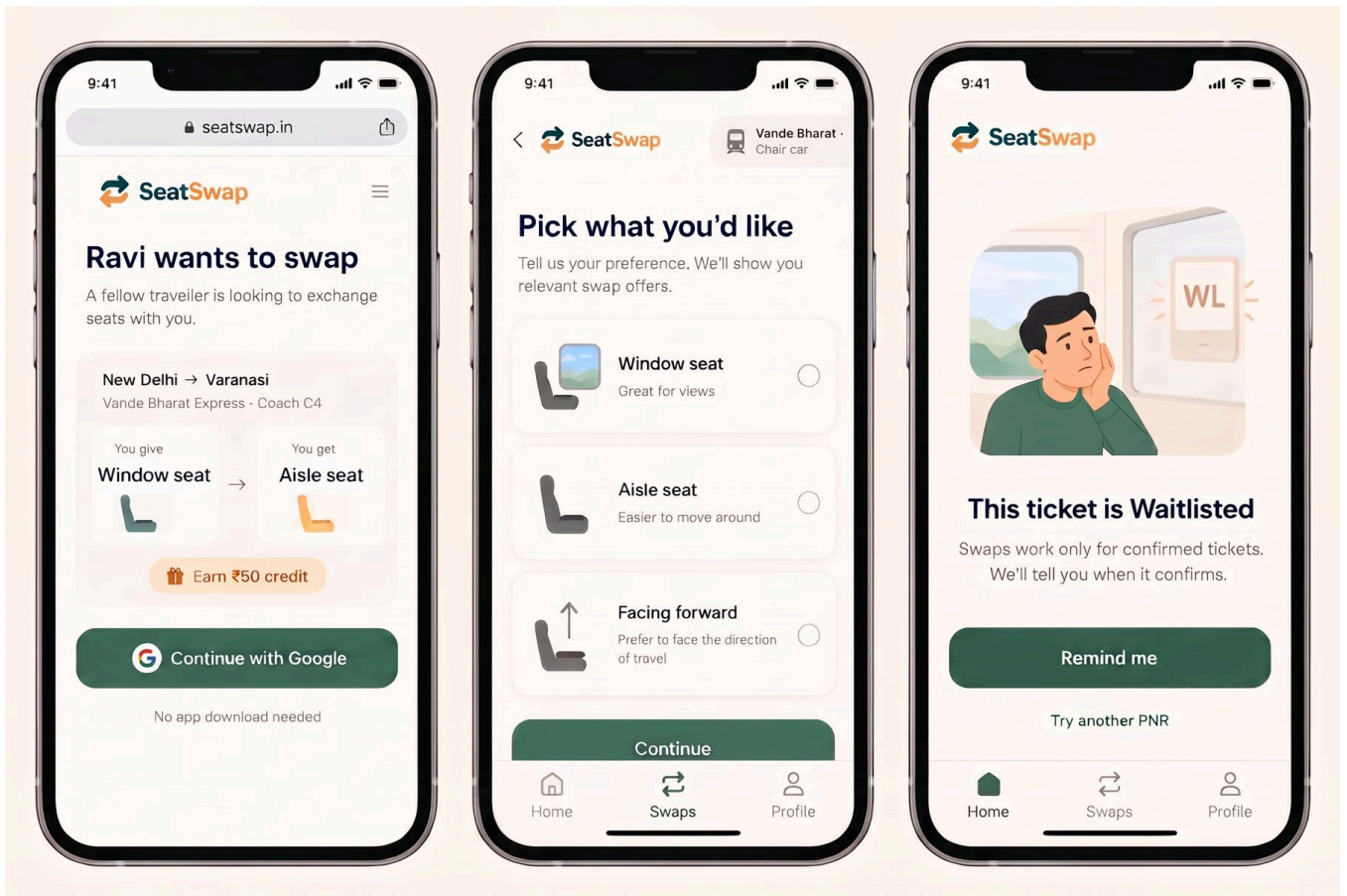
Design 06 Alerts, problems, sharing



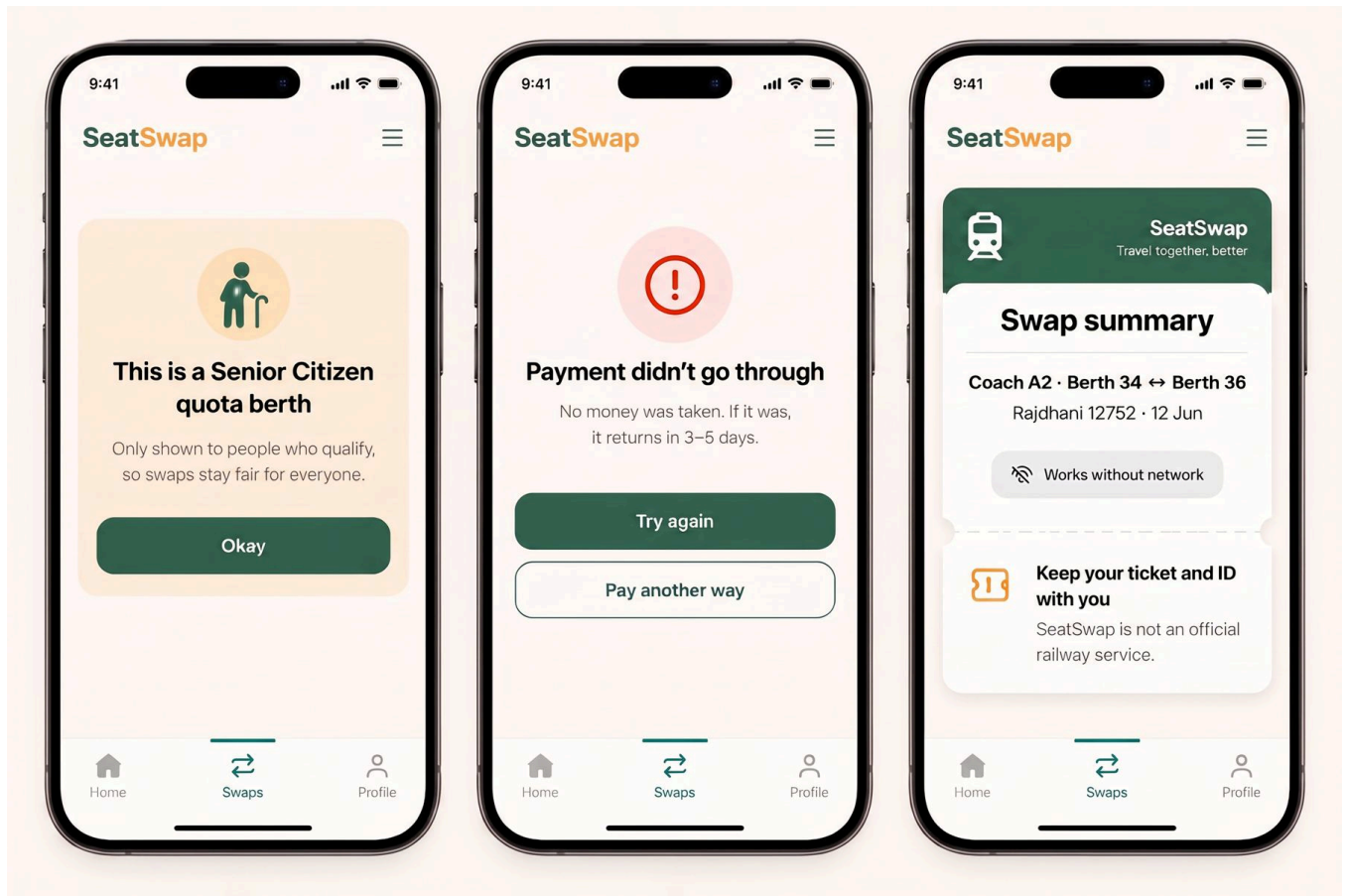
Design 07 On board, first, chart



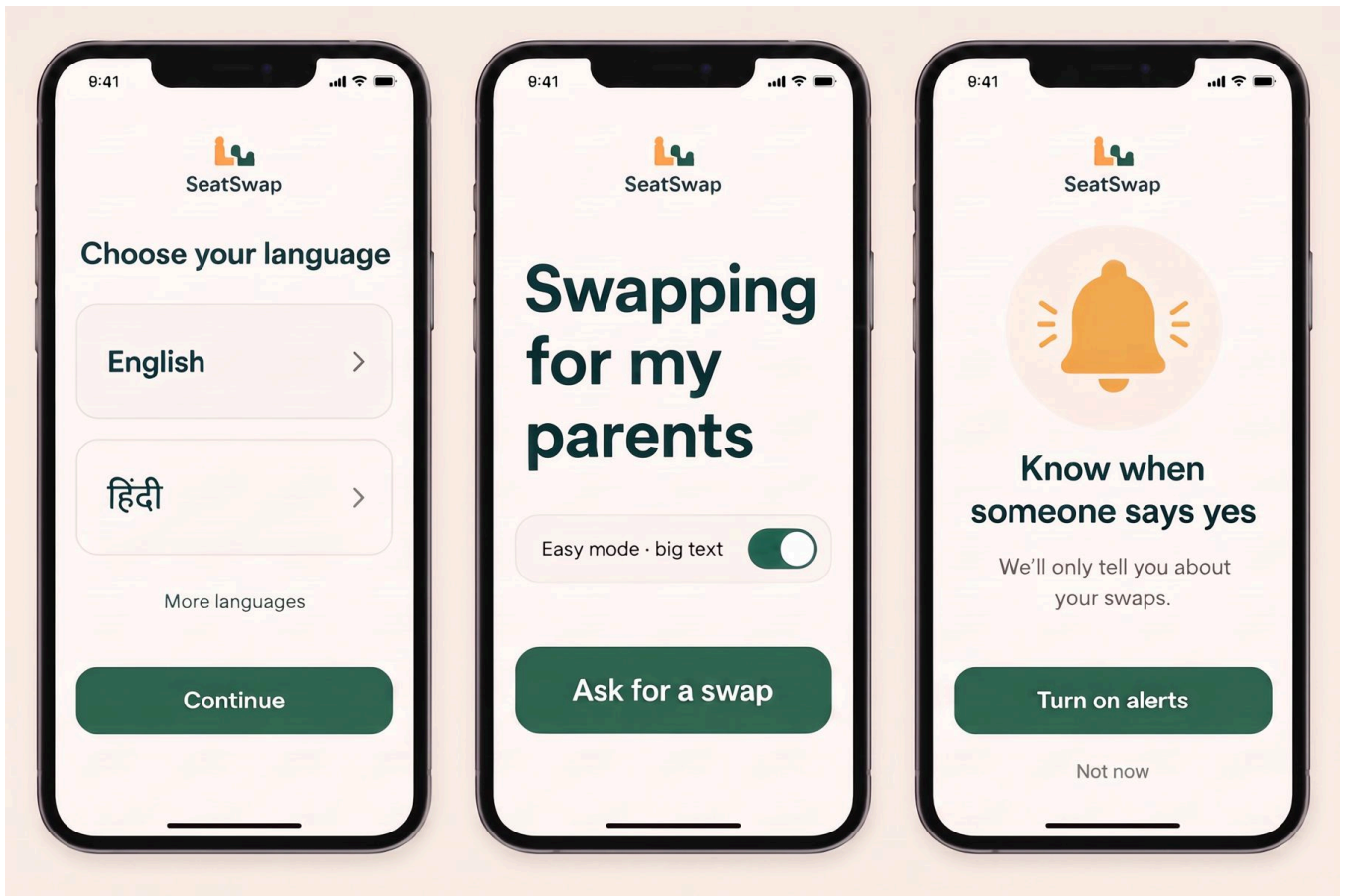
Design 08 WhatsApp accept, chair car, waitlist



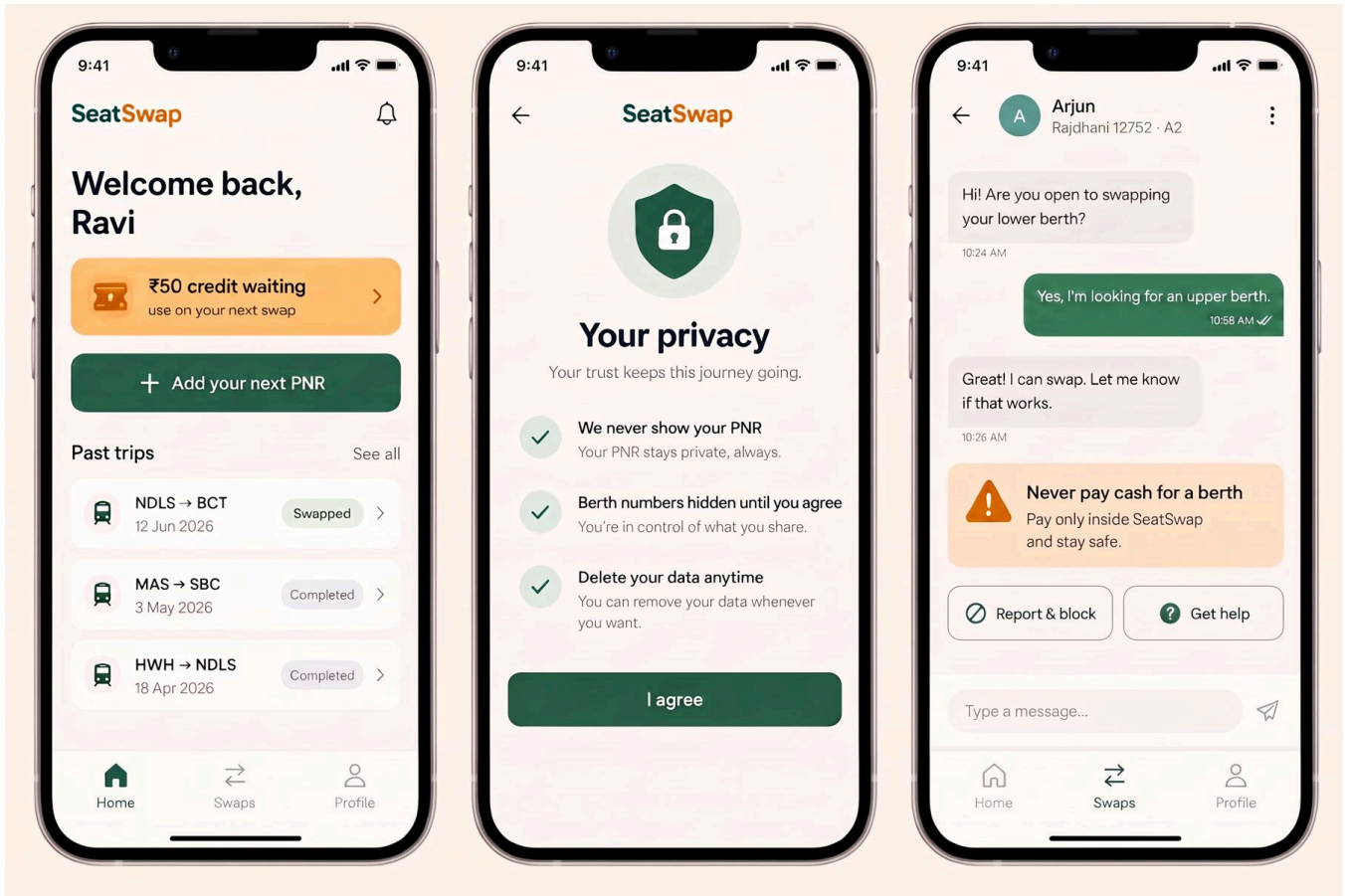
Design 09 Quota, payment, swap summary



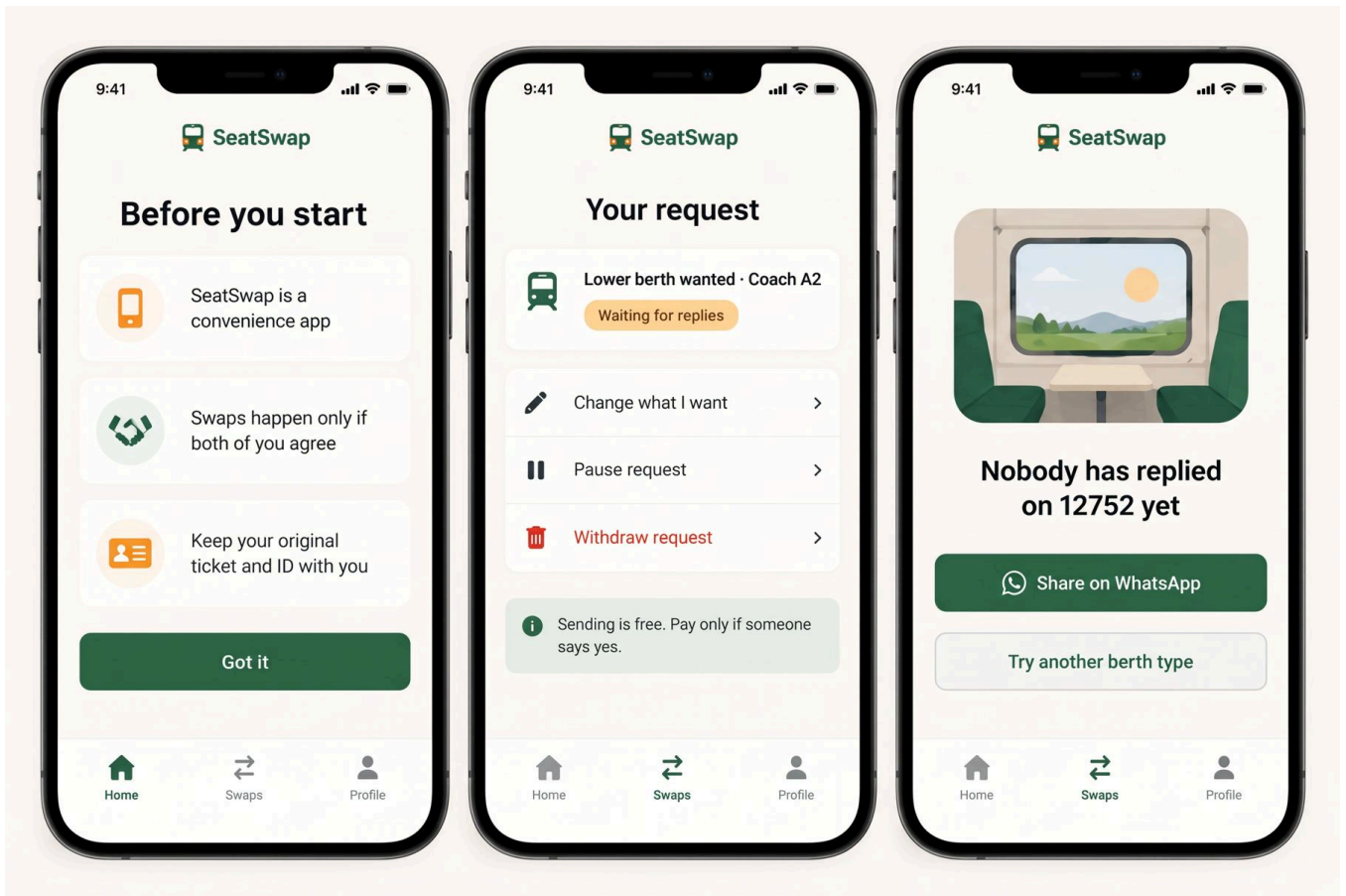
Design 10 Language, easy mode, alerts



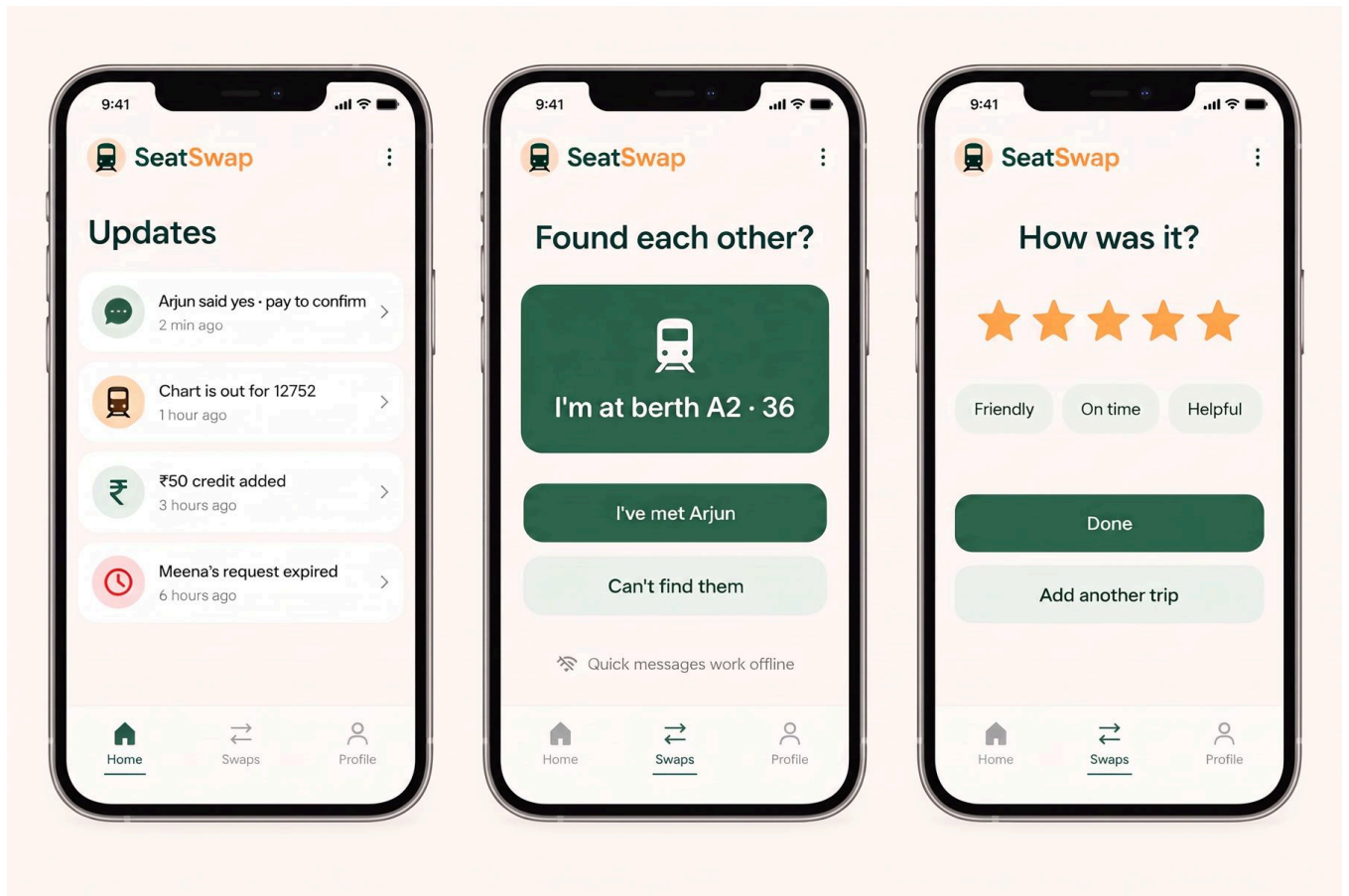
Design 11 Welcome back, privacy, safety



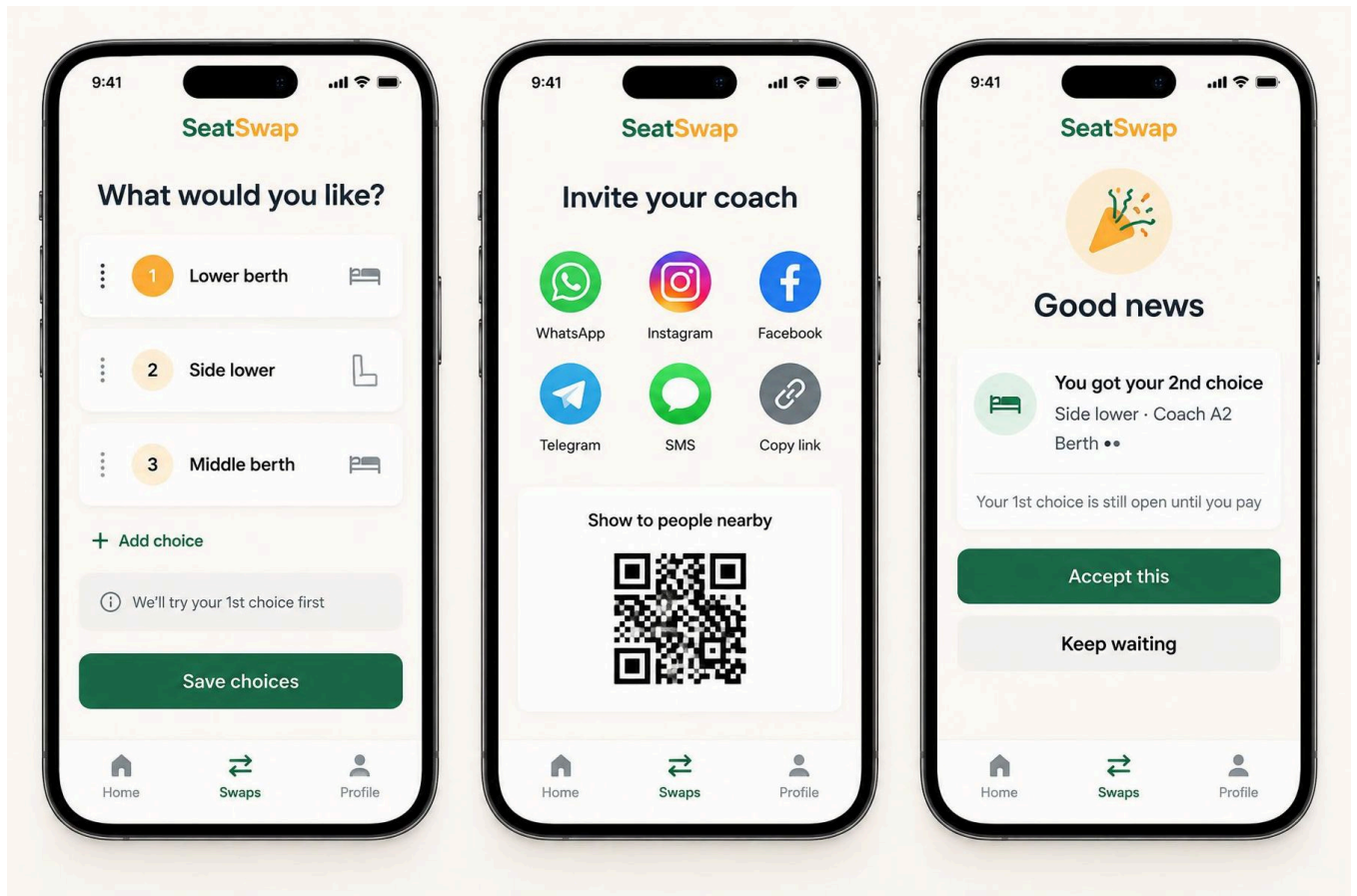
Design 12 Start note, manage request, no reply



Design 13 Updates, meet on board, rating



Design 14 Choices, share anywhere, 2nd choice



Design 15 Admin activity log

 SeatSwap Admin

 Overview

 Activity log

 Users

 Swaps

 Payments

 Credits

 Reports

SAAdmin

Activity log

Track all important events across SeatSwap

AllRequestsPaymentsSwapsReportsSign-ins

Search user, PNR or train

Time	User	Action	Train	Details
22:41	Riya P	 Paid ₹99	12752	UPI
22:40	Arjun S	 Accepted	12752	A2
22:38	Riya P	 Sent request	12752	Wants Lower
22:30	Meena K	 Added PNR	12951	3 passengers
22:22	Karan B	 Reported chat	12752	Cash asked
22:15	Arjun S	 Credit +₹50	12752	Swap done

User timeline: Riya P

22:41 Paid ₹99
Train 12752 · UPI

22:38 Sent request
Train 12752 · Wants Lower

22:35 Chose 1st choice Lower
Train 12752

22:20 Added PNR
Train 12752 · 1 passenger

22:18 Signed in with Google

Export

Design 16 Admin users

SeatSwap Admin

 Overview

 Activity log

 Users

↔ Swaps

Payments

 Credits

Users

Q Search name or email

All

Active today

Reported

Blocked

Name	Joined	Trips	Swaps	Credit	Status
R Riya P	12 Jun	2	1	₹0	Active
A Arjun S	3 Jun	4	3	₹150	Active
M Meena K	1 Jun	1	0	₹0	Active
K Karan B	28 May	3	1	₹50	Reported
V Vikas T	20 May	1	0	₹0	Blocked

A

Arjun S

arjun.s@example.com

Google verified

★ Rating **4.8**

↔ 3 swaps done

 ₹150 credit

[View activity](#)

Block user

Design 17 Admin swaps

 SeatSwap Admin

Overview

Activity log

Users

Swaps

Payments

Credits

Reports

AAdmin

Swaps

All

Waiting

Accepted

Paid

Done

To credit

Swap	Train	Requester	Acceptor	Status	Amount
#1042	12752	Riya P	Arjun S	Paid	₹99
#1041	12951	Meena K	—	Waiting	₹0
#1040	12301	Karan B	Sneha R	Done	₹99
#1039	12627	Rohit M	Anil V	To credit	₹99
#1038	12752	Pooja D	Arjun S	Done	₹49

Swap #1042

Train

12752

Requester

Riya P

Acceptor

Arjun S

Amount

₹99

Timeline

✓

Request sent

22:38

✓

Arjun accepted

22:40

✓

Riya paid ₹99

22:41

Waiting for both to confirm

Move to credit

Mark done

Design 18 Admin payments, credit, reports

↔

SeatSwap

Admin

🏠

Overview

📅

Activity log

👤

Users

↔

Swaps

💳

Payments

📦

Credits

📊

Reports

Payments

Track payments and credits.

💳

Paid today

₹4,950

↶

Moved to credit

₹297

📦

Credit given

₹1,250

💳

Credit used

₹600

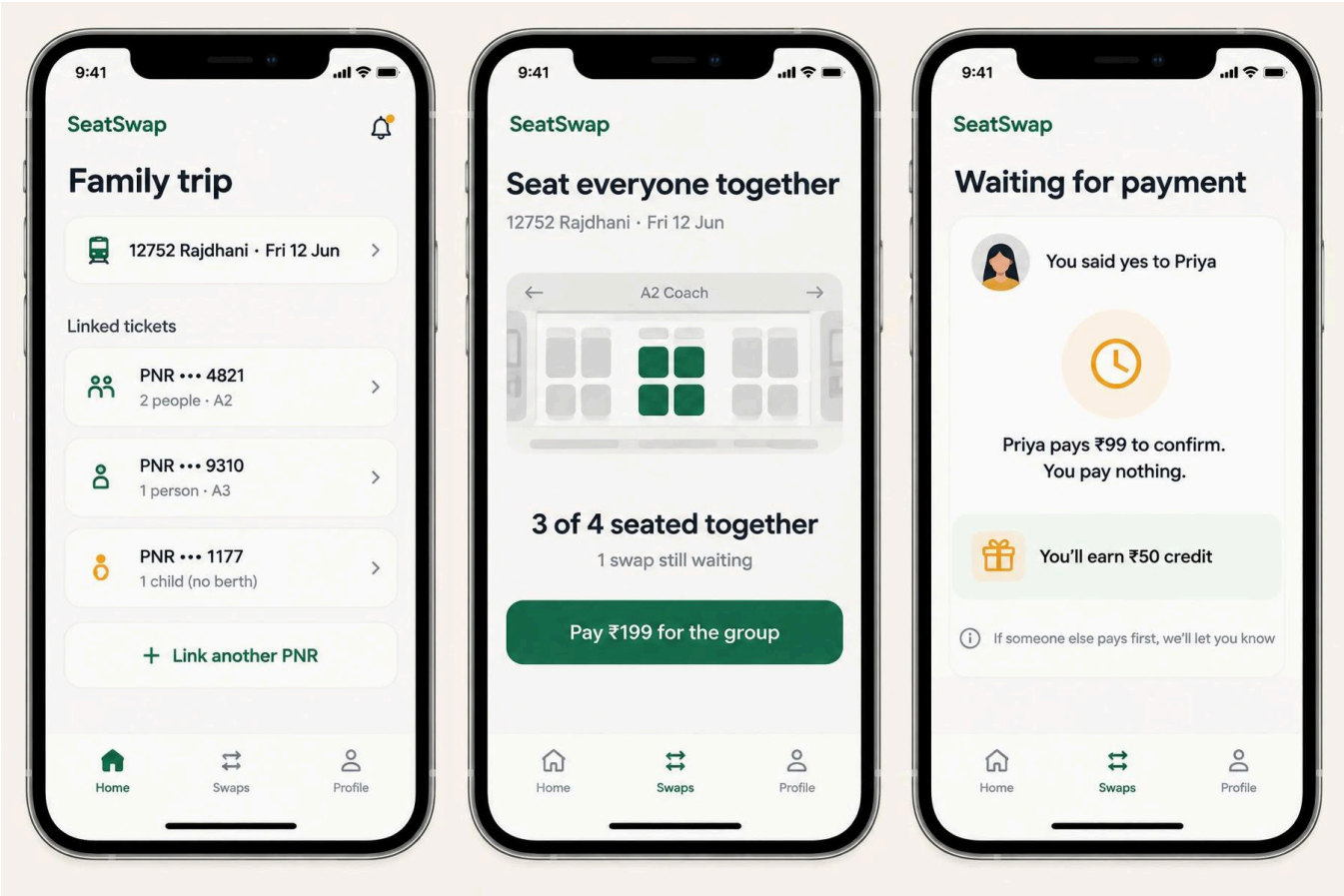
Payments

Time	User	Swap	Amount	Method	Status
22:41	Riya P	#1042	₹99	UPI	Paid
21:12	Pooja D	#1038	₹49	UPI + credit	Paid
20:05	Rohit M	#1039	₹99	Card	To credit

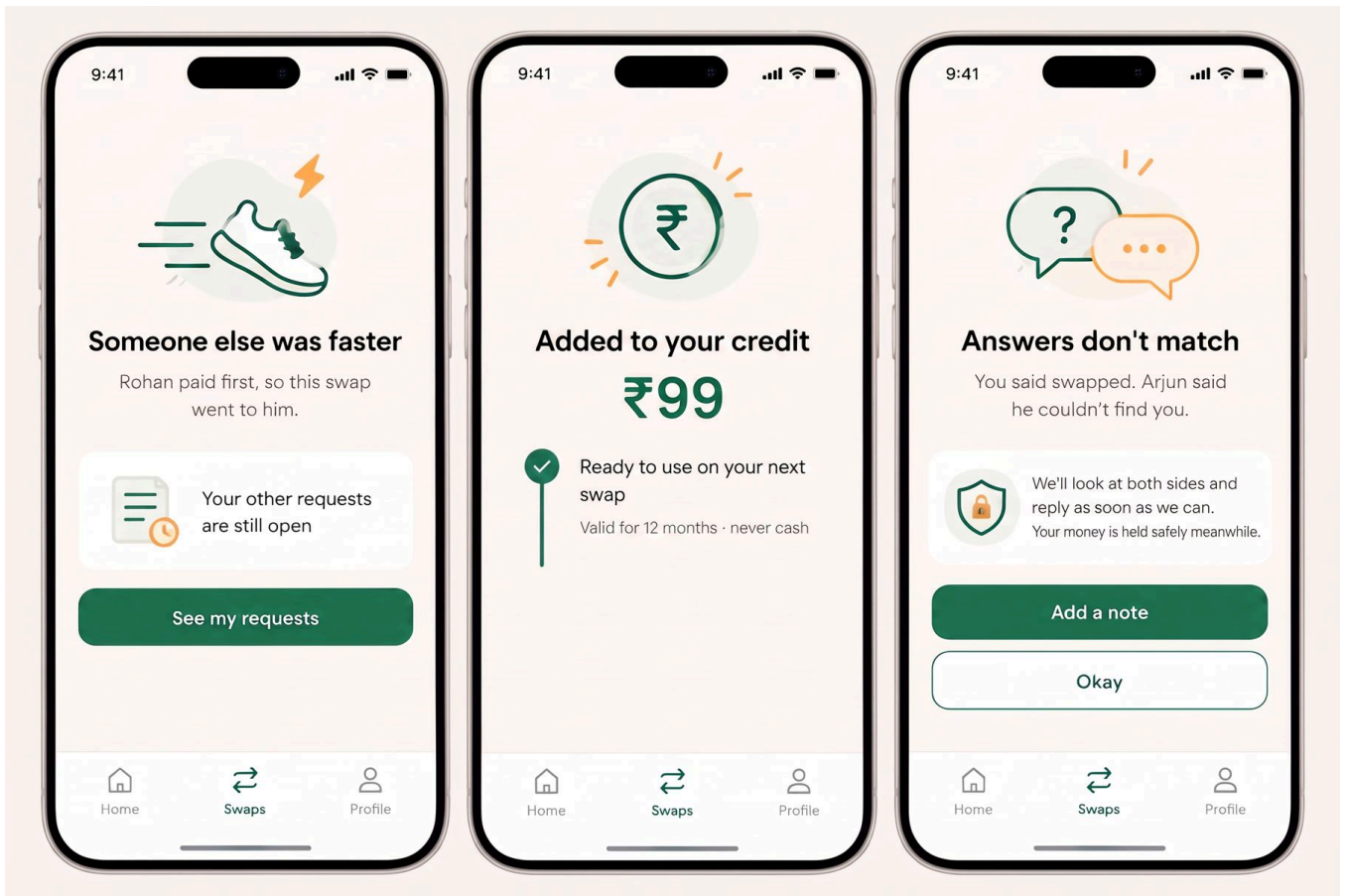
Reports

User	Issue	Status	Actions
Karan B	reported chat · Cash asked	Open	Review Close
Sneha R	Other person didn't come	Open	Review Close

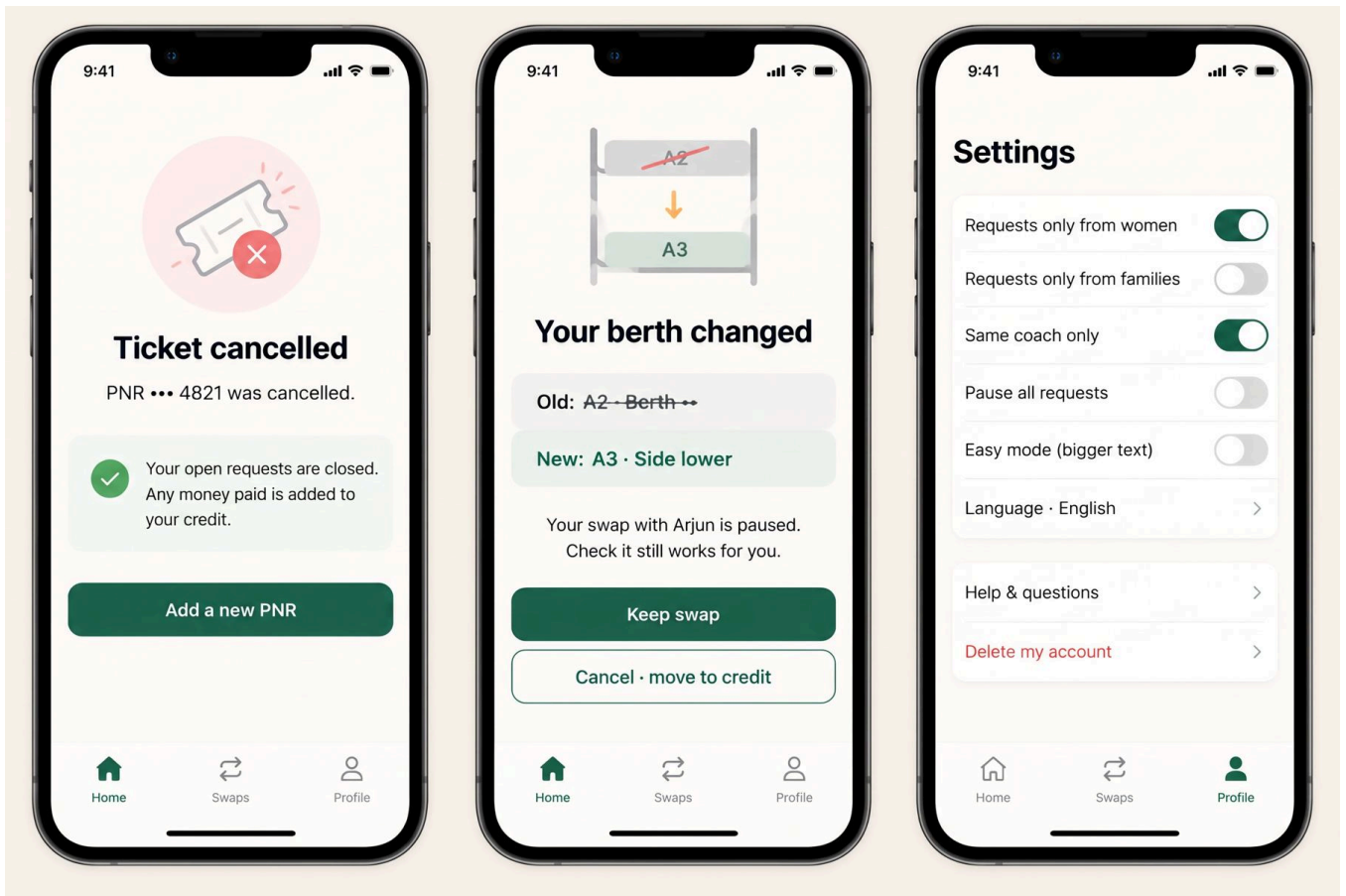
Design 19 Family trip, group, waiting for payment



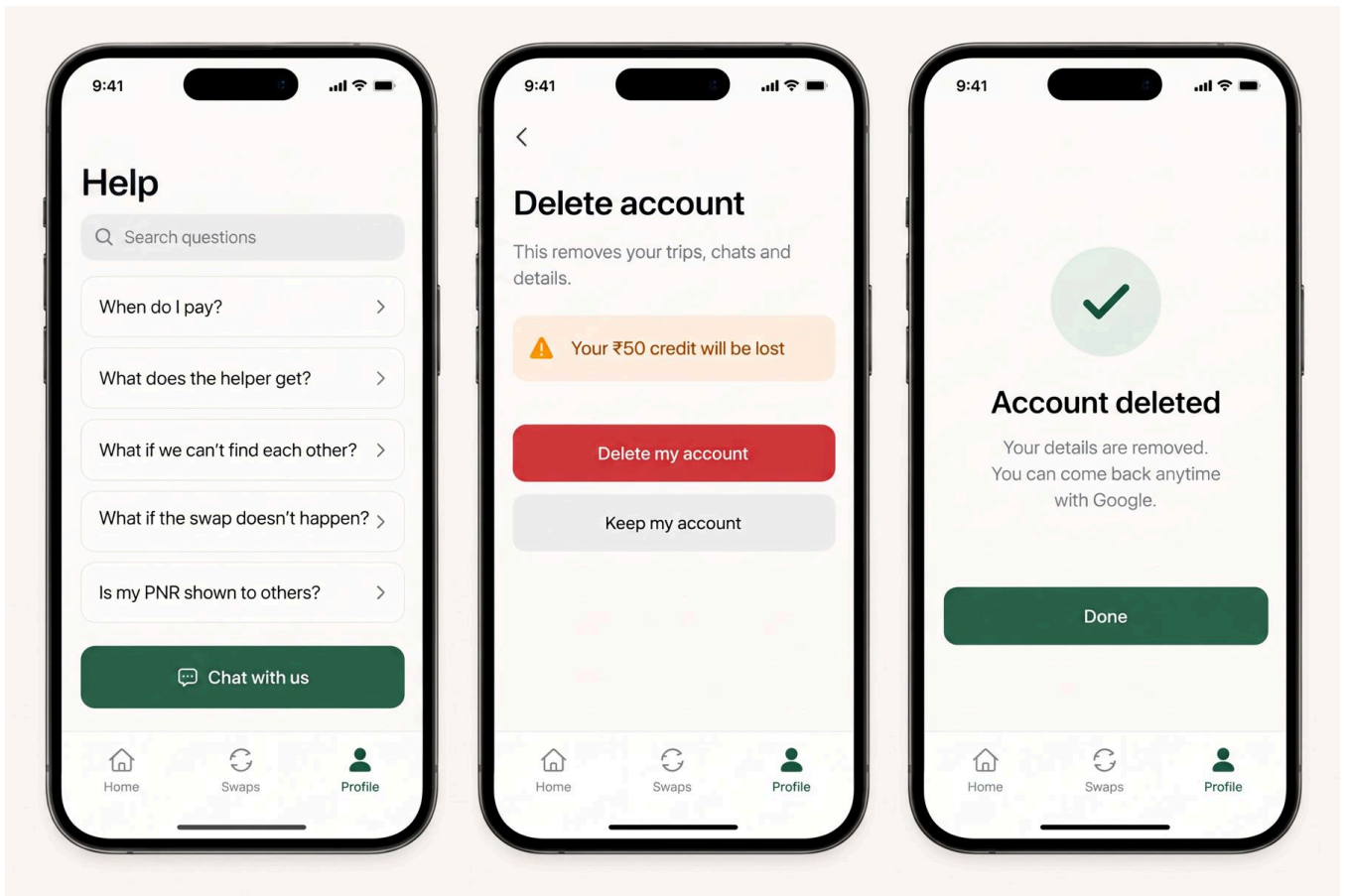
Design 20 Someone faster, credit, answers differ



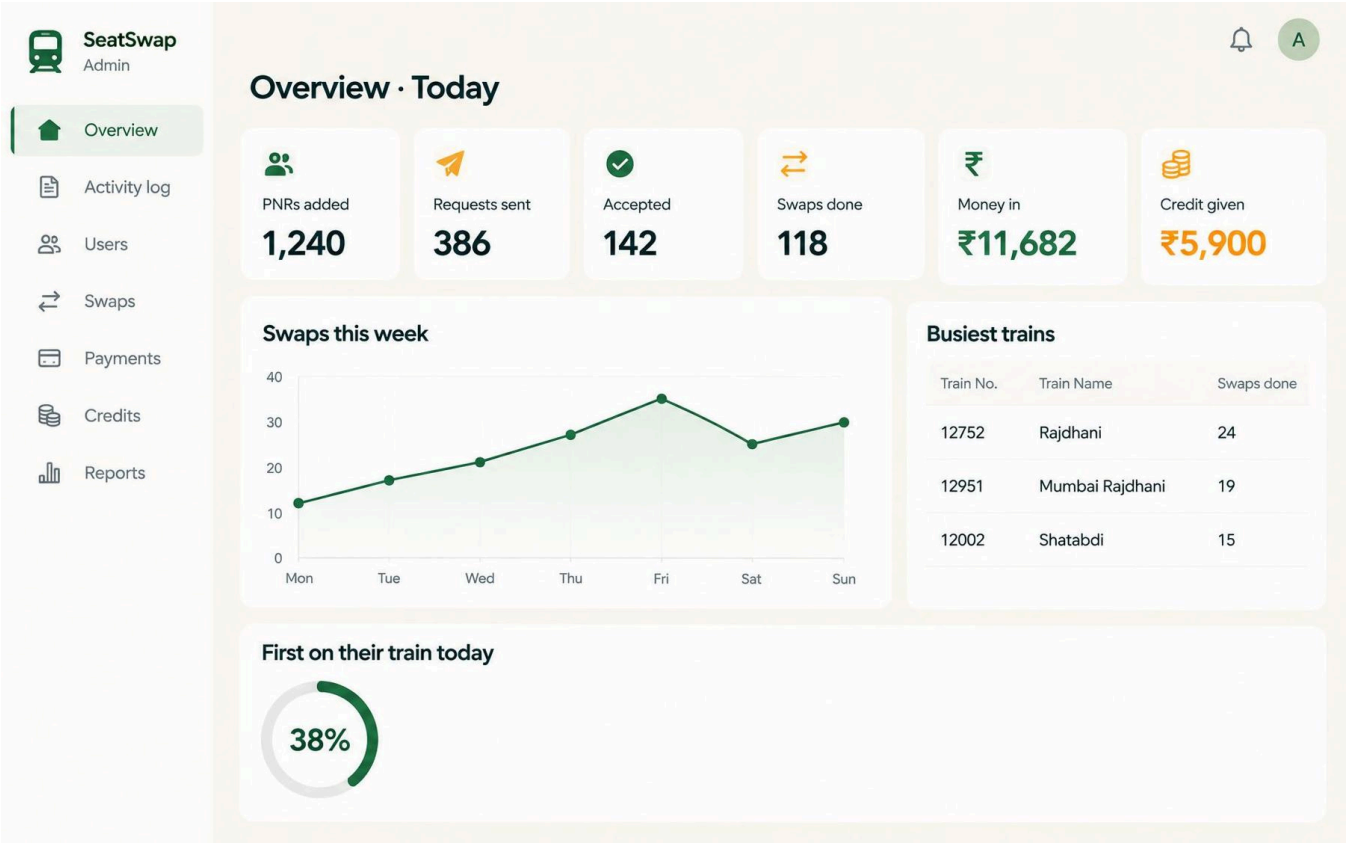
Design 21 PNR cancelled, berth changed, settings



Design 22 Help, delete account, deleted



Design 23 Admin overview



Design 24 Admin credits

SeatSwap Admin

Overview

Activity log

Users

Swaps

Payments

Credits

Reports

Credits

View history

Adjust credit

Credit given

₹5,900

Credit used

₹2,350

Unused balance

₹41,200

Expiring this month

₹1,800

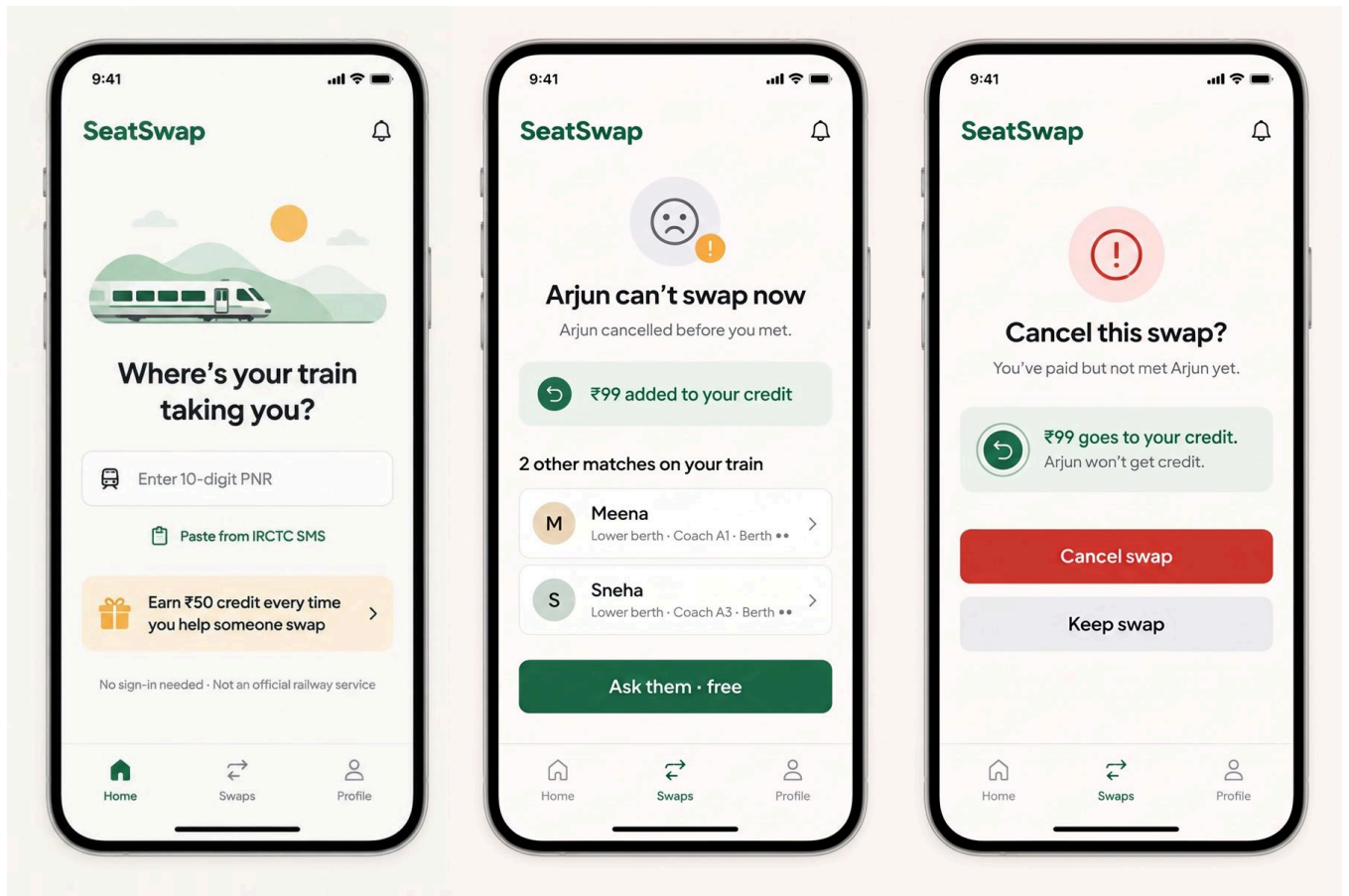
Search user

User	Earned	Used	Balance	Last activity
AS Arjun S	₹150	₹50	₹100	Today
PM Priya M	₹50	₹0	₹50	Yesterday
KD Karan D	₹100	₹100	₹0	3 days ago
MR Meena R	₹50	₹0	₹50	1 week ago

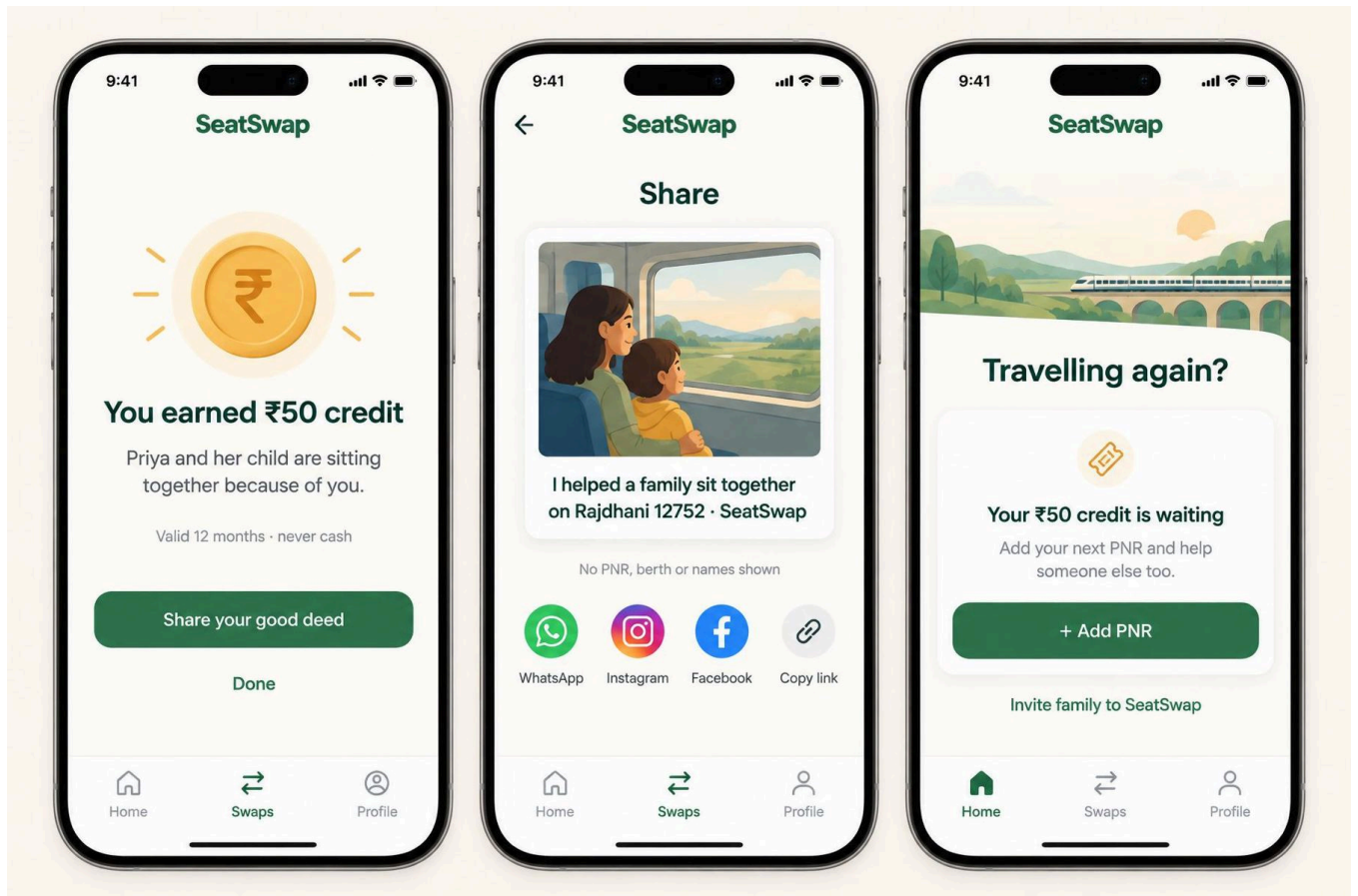
Info

Credit is never cash.

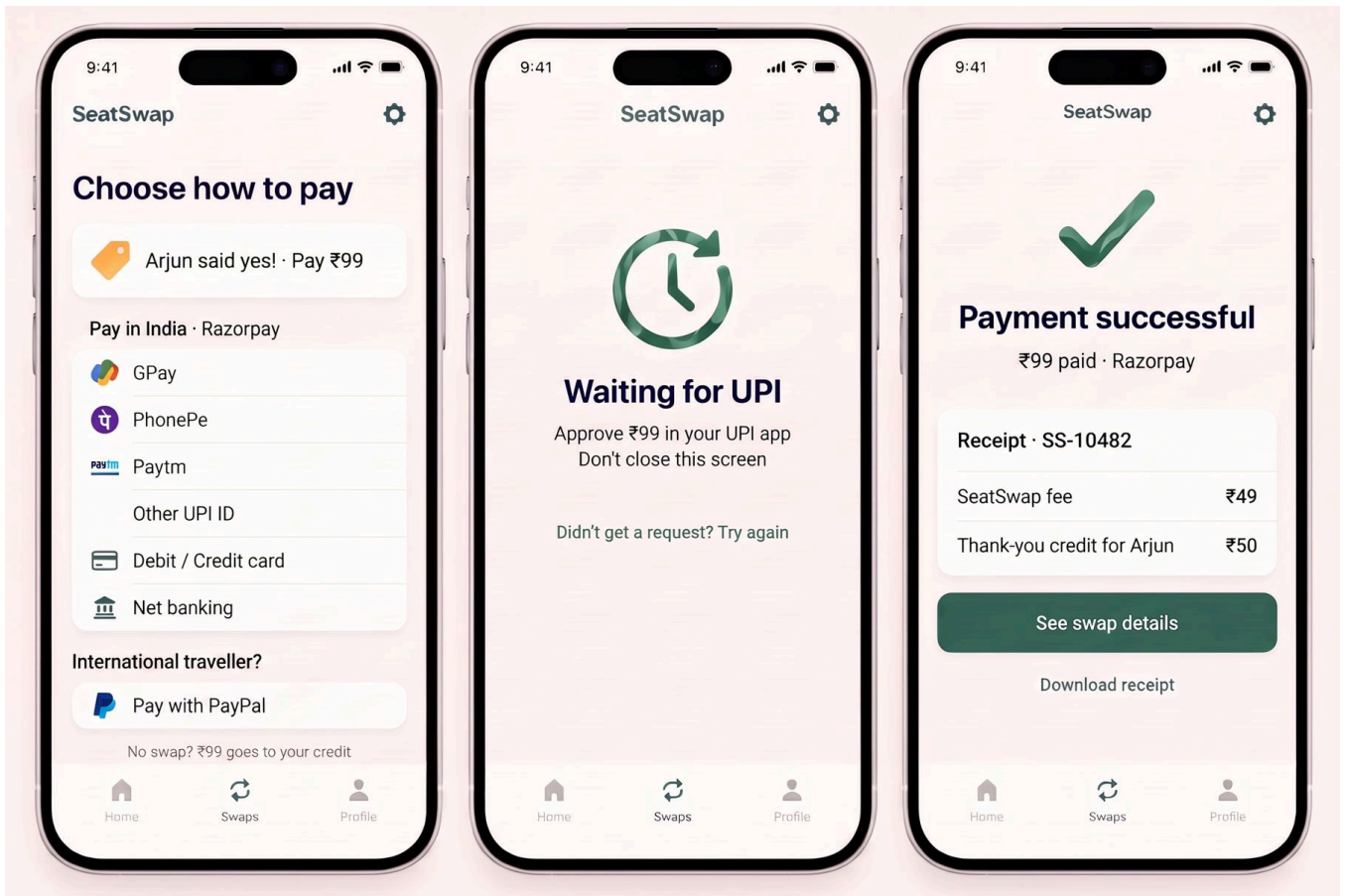
Design 25 First open, partner cancelled, cancel swap



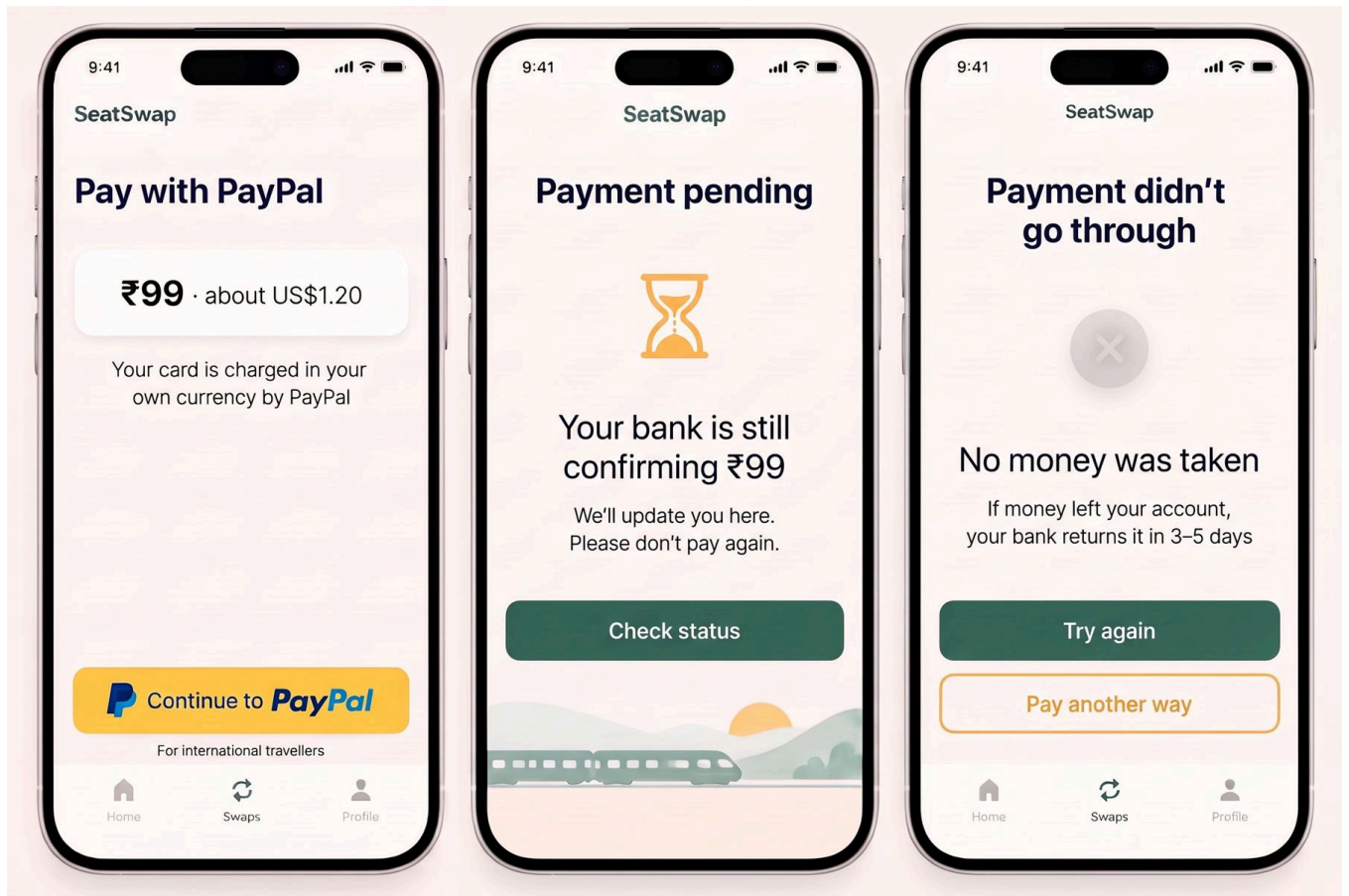
Design 26 Helper earned, share good deed, come back



Design 27 Razorpay pay, UPI waiting, paid



Design 28 PayPal, pending, failed



Design 29 Payments history, receipt, offline

